The Effect of Public Sector Relocations on Regional Development in Germany

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Can governments create private sector jobs in lagging regions by relocating public sector employees? I study a program moving about 5,200 public sector jobs from thriving cities to lagging towns in one of Germany's largest states. Using linked employer-employee social security records, I construct comparison groups outside the policy state by matching municipalities on the indicators that determine program eligibility. Within five years, private employment in receiving municipalities rises by 4.3%, with no offsetting decline in sending municipalities. Following individual public sector workers affected by the relocations, I find that jobs move, but people do not. A quantitative spatial model rationalizes these effects through public-to-private productivity spillovers at receiving locations and the release of floor space for private firms at sending locations. The relocations leave national output and welfare nearly unchanged.

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1 Introduction

Can governments narrow regional disparities in economic opportunities by moving public sector jobs from thriving to lagging regions? Governments have long paid subsidies to firms to locate in targeted lagging regions. Germany, Denmark, France, Canada, South Korea, the United Kingdom, and Sweden are now moving public sector jobs to these regions to stimulate local economic development instead[1]. In 2019, Germany's public wage bill amounted to 8.1% of GDP (Eurostat 2026b), whereas federal and state spending on GRW, the country's main place-based policy program, amounted to 0.03% of GDP (Deutscher Bundestag 2021; Eurostat 2026a). Because political and administrative power has historically been centralized, much of this wage bill is concentrated in national and subnational capitals (Galiani and Kim 2008; Dascher 2000). Relocating just 1% of the public wage bill would move about €2.9 billion a year across space. How redistributing public employment as a place-based policy affects the lagging regions that gain this wage bill and the thriving regions that lose it is therefore a first-order policy question.

My paper answers this question with four contributions, studying an ongoing public employment reallocation program in Germany's second most populous state that is moving 5,196 public sector jobs and 1,170 paid training positions from thriving to lagging regions in two phases from 2015 to 2030. The jobs alone equal roughly 1.7% of the state's pre-policy direct workforce.[2] First, following individual public sector workers, I show that public sector jobs move but public servants largely do not. Few employees of relocated agencies change residence to follow their jobs. After the relocation, agencies are staffed almost entirely by workers from outside the original agency workforce. Second, I estimate the effects of relocating public employment on private employment in regions that lose public jobs and in those that receive them. Within five years, private employment in receiving regions rises by 4.3%, and I find no evidence of offsetting losses in sending regions. Third, I embed a public sector in a quantitative spatial model of Germany, accounting for the office space it occupies in each location. In the German case, the government estimates that the program's first phase alone freed about 37,000 square meters of office space, mainly in the state capital (Bavarian Ministry of Finance 2026). The model distinguishes agglomeration economies within the private sector from productivity spillovers between the public and private sectors. Receiving locations gain public-to-private spillovers as public employment expands. Sending locations lose these spillovers, but the release of floor space allows private firms to expand and can strengthen agglomeration within the private sector. It can simulate local shocks to the public sector, such as the opening, closure, or relocation of agencies, as well as shocks to the private sector, such as firm closures. This makes it a general tool for evaluating place-based policies beyond this application. Fourth, matching the model to the reduced-form private employment responses, I evaluate the repercussions of the policy for the economy as a whole, including regions that neither send nor receive public sector jobs. The estimated

[1] See Freitas (2025) for an overview of past and current public employment relocation programs.

[2] These figures describe intended final staffing. Phase I's scheduled implementation period ended in 2025, although staffing continued to build up. Phase II is still being implemented (Bavarian Ministry of Finance 2026).

public-to-private productivity elasticity is 0.364. The relocations raise private output in receiving counties by 2.8%, while national private output falls by only 0.035% and private workers' expected utility rises by 0.017%. Allocating the same public jobs to destinations with the highest marginal contribution to private output would reduce the national output loss by about two thirds, from 0.035% to 0.012%.

In terms of methods, I combine reduced-form, quasi-experimental evidence with a quantitative spatial model. I use the program's pre-announced eligibility criteria to construct comparison groups. Every municipality within the policy state is benchmarked against the state average on a socioeconomic distress index. The index separates thriving municipalities from lagging ones, which score below 90% of the state average. Lagging municipalities become eligible to receive relocated public agencies. I use administrative data to reproduce the index for all municipalities in Germany outside the policy state. With a matching procedure, I then identify municipalities that would have been eligible for the program had they been located in the policy state. I match the sending municipalities, which the index classifies as thriving, to equally thriving municipalities outside the policy state. This approach yields control groups that are never treated and avoids bias from spatial spillovers of the program.[3] I combine a self-collected dataset of the agencies relocated by the program, the state's official public-agency register with a yearly municipal employment panel that I construct from the Federal Employment Agency's universe of employer-employee matched social security records from 2007 to 2024.[4] With these data, I compare how private employment evolves in sending and receiving treated municipalities and their matches before and after the relocations, assuming that both would have followed parallel trends without the program. The first phase of the program is implemented between 2015 and 2025. To address staggered treatment timing, I use the Sun and Abraham (2021) estimator for the event studies.

A common concern is that place-based policies trade efficiency for equity. When agglomeration economies are similar across regions, the gains from density are largest in productive regions, so moving economic activity toward lagging regions lowers aggregate output (Fajgelbaum and Gaubert 2025). Such moves can raise efficiency only if density also imposes costs, for instance through congestion, or if agglomeration economies are stronger in lagging regions. Reduced-form comparisons between treated and control municipalities cannot tell which case applies, because they miss how the relocations affect regions that neither send nor receive public sector jobs and the economy as a whole. I therefore build a quantitative spatial model of Germany where the government assigns public sector jobs to workplaces, public employees choose where to live, and public agencies occupy commercial floor space that private firms would otherwise use. In sending municipalities, relocating a public sector agency removes the agglomeration economies its jobs generate for nearby private firms, but it also frees floor space that makes room for more private sector agglomeration. Receiving municipalities have a smaller private employment base. In receiving municipalities, the agency adds agglomeration

[3] I additionally exclude municipalities with the strongest commuting links to the policy state from the control pool.

[4] I thank the IAB REGIO group and the IAB RFN Berlin team for their support in accessing the IEB data.

economies between the public and the private sector, but it competes with private firms for commercial floor space.

Tracing individual public sector workers, I find that very few incumbent workers move with their agency, and only 10% commute to the new location. Most of the others either remain in the sending location, typically at another public agency earning higher nominal wages after the relocation, or are no longer observed in the social security records (very likely retirees). At the receiving agencies, almost all workers are new hires from outside the original agency workforce, and many of them previously worked in the public sector. The new hires are more often women, younger, and more educated than the former agency workers, but they earn less. Workforce adjustment appears to occur mainly within the public sector. In 2024, 78% of stayers remain working in the public sector, while 62.4% of new hires had already worked in the public sector in 2014.

My municipal-level reduced-form results show that public employment builds up gradually as the relocated agencies fill their positions. Private employment in receiving municipalities expands alongside public employment. Relative to matched controls, private sector jobs grow up to 4.3% five years after the relocation program starts. Private employment in food and beverage services expand from the start of the program. The public agencies relocated become more female, while the composition of private employment shifts toward men. Commercial rents rise immediately, while residential rents increase later.

In sending municipalities, public employment does not fall immediately. Its estimated response turns negative six years after the first relocation. Private employment estimates are positive in every post-treatment year and reach 2.3% after five years. I therefore find no evidence of private employment losses that offset the gains at the destinations. Knowledge services, particularly business services that serve other firms, contribute to this expansion. Residential and commercial rents also rise despite the release of public office space, as sending locations become more attractive to private firms.

This paper contributes to research on worker adjustment, local employment spillovers, and the spatial allocation of economic activity.

My first contribution is to show how the workforce adjusts when public employers relocate. Gathmann et al. (2020) find that mass layoffs cause persistent local employment losses, while initially employed workers suffer smaller losses; geographic mobility protects younger workers in that initial workforce. I distinguish the adjustment of the original agency workforce from recruitment at receiving agencies. Goldschmidt and Schmieder (2017) find wage losses of about 10–15% after outsourcing, associated with the loss of firm-specific rents. In my setting, average public workplace wages decrease at receiving locations and increase at sending locations. The worker records show that receiving agencies recruit a lower-paid workforce, while many incumbents remain at sending locations and earn higher nominal wages. I follow the workers employed at relocated agencies and show that public sector jobs move but public servants largely do not. Few incumbents change residence, and receiving agencies hire mostly from outside the original workforce.

My second contribution is to the literature on place-based policies, which asks whether spatially targeted interventions improve economic outcomes in lagging regions (Moretti

2012; Kline and Moretti 2014b; Ehrlich and Seidel 2018; Duranton and Venables 2018; Ehrlich and Overman 2020). Most of this work studies subsidies, tax incentives, and infrastructure investment. Relocating public sector jobs differs from these instruments because the government moves its own activity rather than altering private firms' incentives. A small, but related literature studies how public employment affects private employment, using expansions, contractions, and relocations of the public sector (Faggio and Overman 2014; Senftleben-König 2014; Jofre-Monseny et al. 2020; Auricchio et al. 2020; Freitas 2025). Much of the relocation evidence concerns large interventions concentrated in a single destination, such as the German government's moves to Bonn and back to Berlin and the BBC's move from London to Salford (Becker et al. 2021; Faggio, Schlüter, et al. 2025; Nathan et al. 2024). I another study (Freitas 2025), I review of public employment decentralization programs and find gains of about 7 private jobs per 10 relocated public jobs, concentrated in non-tradable activities (ibid.). This paper studies public employment relocation as a place-based policy targeting many lagging locations. It is therefore closer to studies of programs targeting multiple locations in the United Kingdom, South Korea, and Denmark (Faggio 2019; Lee et al. 2024; Jensen et al. 2026). I add evidence from a program whose eligibility rules were announced before implementation, and I estimate effects in the municipalities that send the jobs as well as in the lagging municipalities that receive them. I also contribute evidence on the gendered effects of place-based policies. Women are overrepresented in public employment (Gornick and Jacobs 1998; Gomes and Kuehn 2026) and disproportionately affected when the public sector contracts (Glasmeier and Lee-Chuvala 2011). In receiving municipalities, the relocated agencies become more female, but the additional private jobs that their spillovers create in the private sector go mostly to men. Gender differences in sector specialization therefore relocate with the public agencies.

My third contribution is to bring the public sector into quantitative spatial models (Ahlfeldt, Redding, et al. 2015; Redding and Rossi-Hansberg 2017). Earlier work models public employment in spatial equilibrium in a two-city economic geography model of Bonn (Becker et al. 2021), in a search model of the average Spanish city (Jofre-Monseny et al. 2020), and in models of public sector pay and local public goods (Guillouzouic et al. 2021; Guillouzouic et al. 2024). Rothenberg et al. (2025) study the relocation of Indonesia's capital in a dynamic quantitative spatial model with public employment. I build a quantitative spatial model in which the government assigns public sector jobs to workplaces, public employees choose where to live and commute, and public agencies occupy floor space. A novel feature of the model is that public agencies occupy floor space, allowing me to quantify how their relocation affects real estate markets and private activity. Matching the model to a quasi-experiment , I estimate agglomeration economies between the public and the private sector separately from those within the private sector. In their two-city model of Bonn, Becker et al. (2021) find positive amenity spillovers from public employment, while the fitted productivity spillover to the private sector is close to zero and slightly negative.

My fourth contribution is to the debate on whether place-based policies trade efficiency for equity (Kline and Moretti 2014b; Kline and Moretti 2014a; Busso et al. 2013; Gaubert et al. 2025). In standard spatial models with agglomeration economies, efficiency favors

concentrating activity in productive regions, and moving activity toward lagging regions is justified mainly on equity grounds unless density also creates congestion (Fajgelbaum and Gaubert 2025). Relocating public employment may be a case in which a place-based policy removes a misallocation instead of creating one. Public employment is concentrated not only in capitals but also at their centers. The location of city hall is even a common proxy for a city's central business district (Holian 2019). Political accountability benefits from capitals that are close to their citizens (Campante and Do 2014), but many administrative jobs need neither proximity to political decisions nor proximity to the people they serve.[5] Faggio (2019) describes such jobs as providing tradable public services. They resemble the tradable private sector more than the usual view of the public sector suggests. Yet the literature mostly studies the size of the public sector or the provision of public goods (Alesina et al. 2001; Algan et al. 2002; Guillouzouic et al. 2021). The program I study moves the state's back office, such as tax processing, payroll, land registries, geodata, and IT services, out of thriving centers. These functions serve users statewide without face-to-face contact. Many of these jobs occupy prime locations, the hyper-dense clusters of economic activity where commercial rents and productivity peak (Ahlfeldt, Albers, et al. 2026; Ahlfeldt, Heblich, Seidel, and Yin 2026). By moving these jobs out of prime locations and making room for more productive private firms, the program may correct a historical misallocation of this type of back office public employment. The model shows that the program redistributes activity at a small national output cost, and that destination choice can reduce that cost substantially.

Section 2 describes the policy program and Section 3 the data. Section 4 shows how the workforce adjusts when agencies relocate. Section 5 presents the empirical strategy and Section 6 the effects in receiving and sending municipalities. Section 7 develops the quantitative spatial model and evaluates the relocations for the economy as a whole. Section 8 concludes.

2 Institutional Setting

The public sector is a major employer in Germany and accounts for a substantial share of government expenditure.[6] By choosing where to locate public jobs, governments can direct employment and recurring wage income toward disadvantaged places. Bavaria used this instrument to address spatial inequality within the state.

Bavaria lies in southeastern Germany, bordering Austria and the Czech Republic. Its 13.25 million inhabitants in 2024 made it Germany's second most populous state, with a population comparable to Pennsylvania's (Bavarian Statistical Office 2025a; U.S. Census Bureau 2025). It is also Germany's largest state by area (Bavarian Statistical Office

[5] Studying cuts in Italian public employment, which includes local public good providers such as schools and health services and has long served as a regional subsidy (Alesina et al. 2001), Auricchio et al. (2020) find that an inflated public sector crowds out private employment, mostly in tradable sectors.

[6] On the OECD's comparable general-government measure, employment shares in 2021 were approximately 11% in Germany, 15% in the United States, 21% in France, and 17% in the United Kingdom (OECD 2023, Figure 12.1). In the same country order, the World Bank reports wage bills of 17.4%, 25.8%, 22.1%, and 23.4% of public expenditure in 2019 (World Bank 2024).

2025b).

Economic opportunities differ sharply across Bavaria. Wealthy municipalities around Munich and in the south contrast with struggling areas in Franconia and along the Czech border.[7] The state thus faces a familiar regional-policy problem: how to bring jobs to places that have fallen behind its economic centers. In 2015, Bavaria began reallocating public employment from its major agglomerations to these places as part of its *Heimatstrategie*, a regional development strategy.

To identify eligible destinations, the government benchmarks municipalities across Bavaria against the state average. Municipalities that fall behind on economic and demographic indicators form the eligible set. The government summarizes these indicators in an index covering population forecasts, unemployment, employment density, income, and migration among young adults. A municipality qualifies through either its own index or its district's index, with a threshold of 90 relative to a state average of 100 (Bavarian Ministry of Finance 2015). Panel (a) of Figure 1 shows the municipalities meeting these eligibility criteria. Appendix A reports the index components and their construction.

Within the eligible set, the government favored towns designated as service centers for surrounding communities and places that had not already received substantial support through higher-education initiatives (Bavarian Ministry of Finance 2025). Ministries assessed whether potential destinations offered suitable premises and infrastructure for the agencies they proposed to relocate. They selected sites in coordination with the finance ministry; the cabinet approved the program, and the responsible ministries implemented it (Kaltenhauser 2023). Panel (b) adds the service-center designations, while panel (c) shows the selected destinations and origins. Some selected destinations lie outside the index-defined eligible set.

The agencies selected for relocation perform functions that are not tied to local public good provision. Tax processing, information technology, and other specialized administrative functions can serve the same users from a different municipality. Ordinary schools and local police stations are therefore excluded from the relocation program (ibid.). The program includes whole and partial agency moves, as well as the establishment or expansion of offices. For agency moves, Munich is an important origin: relocating an office shifts public jobs away from the state's largest employment center while adding them to a receiving town.

I study both phases of the public employment reallocation program. Together, they envisage 5,196 jobs and 1,170 paid training positions, equivalent to about 1.7% of Bavaria's pre-policy direct state workforce when counting jobs alone.[8] The training positions include public-service apprenticeships organized as degree programs: participants receive remuneration during their training, so these positions also bring wage income to receiving

[7] Before the program, average annual wages in high-wage municipalities, concentrated around Munich and Starnberg, were more than twice those in low-wage municipalities, concentrated in Upper Franconia and Lower Bavaria (Bavarian Statistical Office 2015).

[8] The government's March 2026 statement reports intended final staffing of 2,511 jobs and 730 training positions in phase I, and 2,685 jobs and 440 training positions in phase II. Phase II remains under way, with implementation extending to 2030 (Bavarian Ministry of Finance 2026). The employment benchmark uses the direct state workforce before the program; Appendix A gives the calculation.

places (Freistaat Bayern 2026, Art. 75). I assembled a relocation dataset that records agencies, destinations, origins, staffing, and implementation dates. Appendix A documents the program's administrative components and intended final staffing.

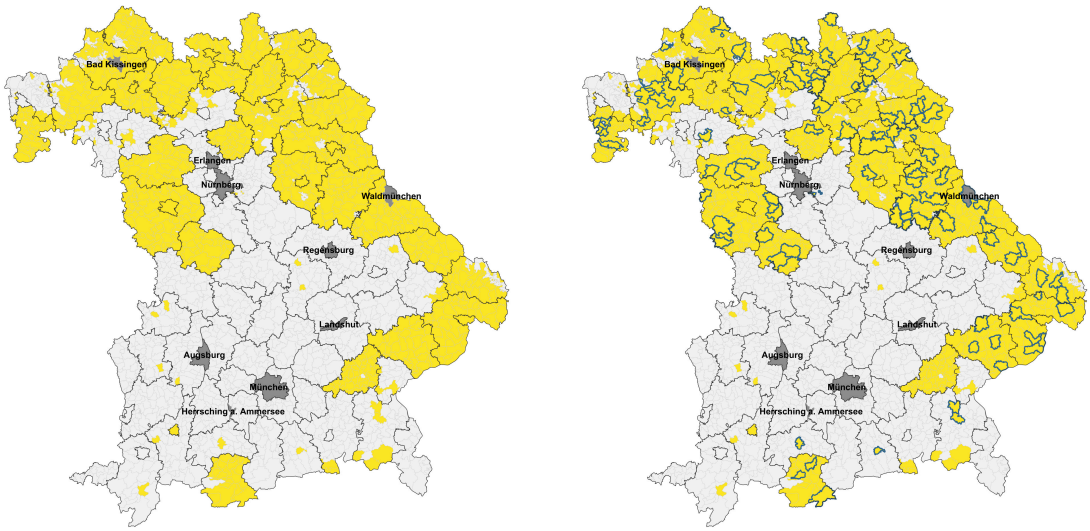
Most workers in Bavaria's direct state administration are civil servants. Before the program, civil servants and judges accounted for 68.4% of this workforce; the remaining workers held employment contracts (Statistical Offices 2014; Bavarian Statistical Office 2014). Civil servants have a distinct legal status and are absent from the social-security employment records used in the empirical analysis. Municipal civil-service employment data from the statistical offices will allow me to extend the analysis to both groups of public employees.

Public base pay does not vary across Bavarian municipalities for a given position and experience level. Contractual employees are paid under the collective agreement for state-government employees, known as TV-L, which Bavaria negotiates jointly with the other participating German states. Wage setting is thus coordinated nationally rather than determined by individual agencies or municipalities. Civil-service pay is set by Bavarian law. Both systems use schedules in which the job determines the pay grade and experience determines progression within it, much as grades and steps organize pay in the US federal General Schedule (TdL 2025; Freistaat Bayern 2026; OPM 2026).

The schedules provide a benchmark for earnings in the relocated jobs. Under the February 2025 schedule, qualified administrative work accessible without a university degree, illustrated by grade E9a, starts at about €3,520 gross per month. A position requiring a master's-level qualification, illustrated by E13, starts at about €4,630 (LfF 2025b). Appendix Table 6 gives the corresponding civil-service grades and pay ranges, distinguishing career level from experience.[9]

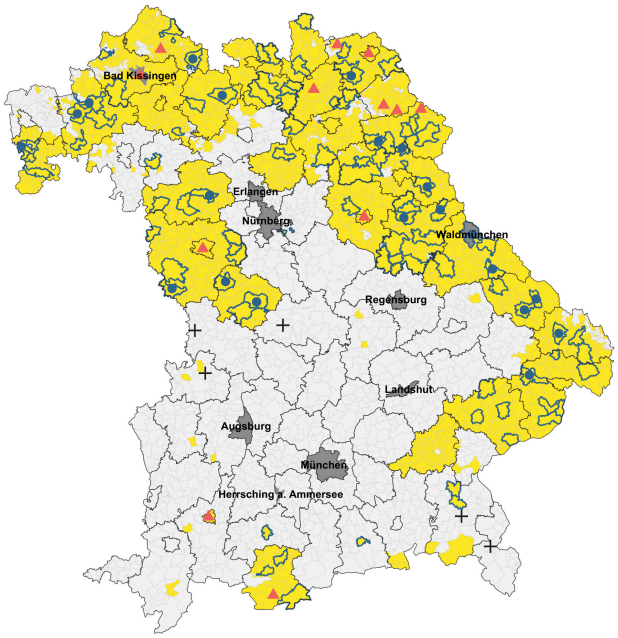
Agencies build up employment at their destinations gradually. An office may open with a small team, recruit additional staff, and move into permanent premises only later. Positions can shift as workers leave or retire at the origin, allowing employment at the destination to grow over several years (Bavarian Ministry of Finance 2026). I date treatment to the government press release announcing an office's opening. This convention captures responses that occur while staffing is still being built up, and the event studies allow me to check for responses before the announced opening.

[9] A public-sector wage premium also depends on the private-sector alternative. Using German survey data, Bonaccolto-Töpfer et al. (2022) find a wage advantage for women but a pay penalty for men after accounting for worker characteristics and selection. These estimates refer to Germany as a whole.



(a) Municipal eligibility

(b) Service-center designations



(c) Receiving and sending locations

Figure 1. Geographical targeting and site selection. Panel (a) marks municipalities meeting the index-based eligibility criteria in yellow. Panel (b) outlines eligible service centers in teal. Panel (c) adds receiving municipalities: blue circles mark service centers, red triangles other municipalities, and black crosses municipalities outside the index-defined eligible set. Sending locations appear in gray throughout.

Employees do not have to move when their agency relocates. The personnel framework rules out compulsory permanent transfers, relocation-related dismissal, and involuntary downgrading. Workers can seek reassignment, while those who follow an agency can receive training and support for moving or commuting, including a conditional €3,000 mobility payment. Telework is available where duties permit (Bavarian Ministry of Finance 2021); in practice, two work-from-home days per week are common. Incumbents can therefore stay, commute, or move, while agencies can also recruit at the destination. Public jobs therefore need not bring an equal number of new residents. Section 4 analyzes these worker responses.

Relocations also change the use of office space. Receiving agencies rent, adapt, or construct premises, often before reaching full staffing. At the origins, the government estimates that phase I freed about 37,000 square meters of office space, mainly in Munich (Bavarian Ministry of Finance 2026). Vacated premises create scope for other uses, including private-sector offices. Sending locations thus lose public jobs while potentially releasing space for other employers. Section 6 examines employment responses at both ends of the relocations.

Providing new premises can be a substantial part of relocation costs. For three completed tax-office relocations with matched cost and staffing figures, construction spending averaged about €180,000 per public job (Kaltenhauser 2023). Firm-specific investment incentives provide a benchmark: approved direct grants for Northvolt's Heide battery plant amounted to about €230,000 per planned job, while proposed support for Intel's Magdeburg semiconductor plant amounted to €3.3 million per planned job.[10]

3 Data

Studying public employment relocation requires knowing where agencies move and what happens to the people and firms around them. I assemble a record of the Bavarian program, use successive editions of the Bavarian public-agency register to document agency locations, and measure employment with linked employer–employee data from the Institute for Employment Research (IAB). The empirical analysis follows municipalities over time. I also construct county-level measures for the quantitative model, which I quantify at the county level.

3.1 Data on the Relocation Program

I construct the relocation dataset by combining official program records. Starting from the BayernAtlas records, I collect the agency name, responsible ministry, origin and destination, intended staffing, program phase, and implementation status (State Office for Digitization, Broadband and Geodetic Measurement 2024). I then follow each project through the annual *Heimatstrategie* reports, government announcements, agency websites,

[10]The relocation ratio covers construction spending per public job accommodated. The Northvolt and Intel ratios use announced support and planned direct employment; Intel cancelled its project in July 2025 (Intel 2025). These figures illustrate spending magnitudes, not relative cost-effectiveness. Appendix A documents the calculations and other reported relocation costs.

and local newspaper coverage. These sources turn a list of planned relocations into a record of where the policy was implemented and when employment could begin to change.

Implementation often unfolds over several years. An agency may open a branch before moving into its permanent building, while staffing grows toward the announced target. I therefore distinguish the start of operations from construction milestones and intended final staffing. I use the official state government sources to establish implementation dates. The program totals in the preceding section describe the intended scale of program rollout phases I and II.

I complement these project histories with a new source for studying the geography of public employment: the Bavarian public-agency register (Bavarian State Statistical Office 2026). Its records identify agencies and their addresses, administrative affiliations, municipality codes, and headquarters or branch status. I reconcile the January 2014, March 2023, and June 2026 editions to follow agencies across changes in names, identifiers, and addresses. Comparing these records with the policy histories helps distinguish a move to another municipality from an address correction or an organizational change. It also reveals why a partial relocation may leave the parent agency at its original address. The register and IAB data serve separate purposes: the former documents agency locations, while the latter measures employment at the establishment level. Appendix B describes the registry reconciliation.

3.2 Employment Data

The main employment source is the IAB's Integrated Employment Biographies (IEB), from which I use linked worker and establishment records for 2007–2024. The IEB combine employers' social-security notifications with administrative records on benefit receipt, job search, and labor-market programs (IAB FDZ 2026; IAB 2017). This source offers a combination essential for the analysis: broad coverage of insured employment, detailed locations, and worker histories that continue across employers. It allows me to study small local labor markets and to connect changes in local employment with the movements of individual workers.

The records distinguish the municipality where a person works from the municipality where that person lives. I use this distinction to construct annual workplace and residence employment measures and to observe commuting across municipal boundaries. The same person and establishment identifiers link employment spells over time; the records also report industry, earnings, education, and demographic characteristics. These dimensions make it possible to ask whether employment grows around a receiving agency, whether local residents benefit, and whether the adjustment involves changes in residence or commuting. County-level aggregates provide the corresponding measures for the model comparison.

Table 1 describes employment growth and the administration share in receiving and sending municipalities. Workplace employment grew by 6.0% in receiving municipalities and 14.9% in sending municipalities between 2014 and 2024; employment among residents grew by 10.4% and 15.8%, respectively.

Industry and employment status answer different questions. Industry codes identify

	Receiving municipalities		Sending municipalities	
	2014	2024	2014	2024
Workplace employment (2014 = 100)	100	106.0	100	114.9
Administration workplace employment (2014 = 100)	100	120.0	100	122.8
Other-industry workplace employment (2014 = 100)	100	105.3	100	114.6
Employed residents (2014 = 100)	100	110.4	100	115.8
Administration share of workplace employment (%)	4.68	5.27	4.24	4.58
Establishments	715	659	14,742	14,213
Municipalities	34	34	6	6

Table 1. Employment growth and composition in treated municipalities. Employment indices set the corresponding 2014 mean to 100. Administration shares are unweighted mean municipal percentages; establishments are unweighted municipal means. Administration denotes WZ 2008 division 84.

public administration, defense, and social security, but do not identify every publicly owned employer. Public schools and hospitals, for example, fall in other industries. Civil servants are also outside the IEB employment records because their jobs are exempt from social-security contributions, so relocated civil-service positions are not observed in the employment data. Self-employment is not covered either.

The worker histories also provide evidence on how agencies adjust their workforce. I identify candidate establishments in two steps. First, I use the workplace postcodes recorded for program origins and destinations to locate public-administration establishments in the IAB data. Second, I look for establishment openings, closures, and unusually large employment changes between 2014 and 2024. At destinations, I retain establishments that are absent in 2014 but present in 2024, or that expand by at least two baseline standard deviations. At origins, I retain establishments that disappear by 2024 or contract by at least one baseline standard deviation. The baseline standard deviation is calculated across establishments' employment changes from 2010 to 2013, separately for origin and destination postcode areas. This procedure captures both whole-establishment changes and substantial expansions or contractions within continuing establishments.

I then use worker identifiers to follow employees of candidate sending establishments in 2014 through 2024. Among those subsequently employed at candidate receiving establishments, changes in recorded residence distinguish movers from workers who keep their residence and commute. I also identify employees at receiving establishments who were not in the original sending cohort. Section 4 presents this evidence. Appendix B gives the sample definitions, including the treatment of postcodes that occur at both ends of the program.

The model uses employment from the 2011 German Census. Its reference date, May 9, 2011, precedes the program, and its employment coverage includes civil servants as well as workers outside social-security employment (Statistical Offices' Research Data Centres 2026). The Census provides workplace and residence counts by industry at the

county level. It supplies the baseline cross-section for the model, while the annual IAB panel supplies the empirical changes.

Table 2 reports the administration shares of county employment in the 2011 Census. Administration accounts for an average of 6.9% of workplace employment across German counties and 6.0% in the policy state. Workplace and residence measures describe different sides of local labor markets and provide the baseline for the model.

	Germany		Bavaria	
	Mean	SD	Mean	SD
Administration share of workplace employment (%)	6.9	3.0	6.0	2.8
Administration share of employed residents (%)	7.2	2.0	6.3	1.4
Counties	412		96	

Table 2. Administration shares of county employment in the 2011 Census. Unweighted means and sample standard deviations across counties in the published source files, including independent cities. Sector shares are percentages calculated within each county. Public administration comprises administration, defense, and compulsory social security (WZ 2008 section O). Source: 2011 Census tables 2000X-2041 and 2000X-2045; Appendix B describes the extraction.

3.3 Controls Used in Mahalanobis Matching

For matching, I assemble the indicators used to assess regional disadvantage: the population outlook, unemployment, workplace employment density, income per taxpayer, and net migration of young adults. I use administrative data from the German Regional Database and Federal Employment Agency, together with the government's index documentation, to reproduce the policy benchmark outside Bavaria (German Statistical Offices 2026). I also account for municipal size and pre-policy outcome trajectories.

3.4 Travel Costs

Commuting connects the local effects of the policy across space. I use Federal Employment Agency origin–destination flows to measure these connections and road travel times to characterize the cost of reaching a workplace. For the county model, the travel-time matrix describes journeys between county locations. I compute road travel times with OpenStreetMap routing through OSRM.

3.5 Rents, Housing Prices, and Commercial Floor Space

To measure the price of residential space, I use the micro-geographic rent indices of Ahlfeldt, Heblich, and Seidel (2023). I take the municipal rent measures for 2011 and aggregate them to counties using population weights for the same year. These inputs describe the housing costs households face in the model and allow location choices to depend on more than wages and commuting.

Office rents provide a separate measure of the cost of space used by firms. I use the municipal office-rent series associated with Ahlfeldt, Heblich, Seidel, and Yin (2026) and construct corresponding population-weighted county measures for 2011.

4 Worker Adjustment: Descriptive Evidence

Public sector jobs move, but public servants largely do not. Few workers change residence and take jobs at receiving agencies. Most stay in the sending locations and continue working at another public sector agency. Others disappear from the employment records and are very likely retirees. Receiving agencies recruit almost their entire workforce from outside the original sending workforce. This section follows the original workers and examines who fills the jobs at the destinations.

I use fine-grained sector and *postcode* matching to identify public sector establishments affected by the policy, then follow the 2014 workforce of those that close or substantially contract in sending locations. I call these workers *incumbents*. The matching combines public sector industry codes with the workplace postcodes recorded for the program's origins and destinations. Section 3 and Appendix B describe how openings, closures, and substantial employment changes identify the establishments used in the analysis.

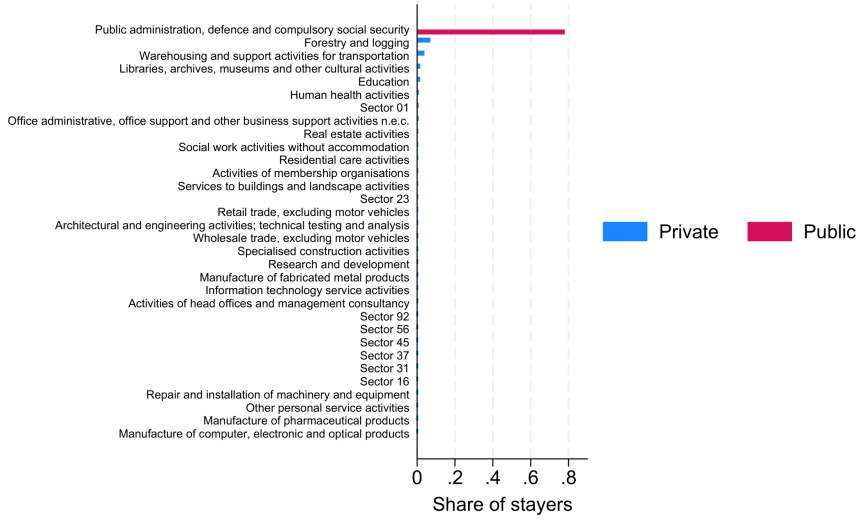
I examine where incumbents work and live in 2024. Each outcome group is expressed as a share of the original 2014 workforce. I then turn to establishments that open or substantially expand in receiving postcodes and examine who works there in 2024. Recruitment shares use this receiving workforce as their denominator. These comparisons answer different questions. The first follows the original workers. The second describes the workforce at the destinations.

4.1 Adjustment of the Original Workforce

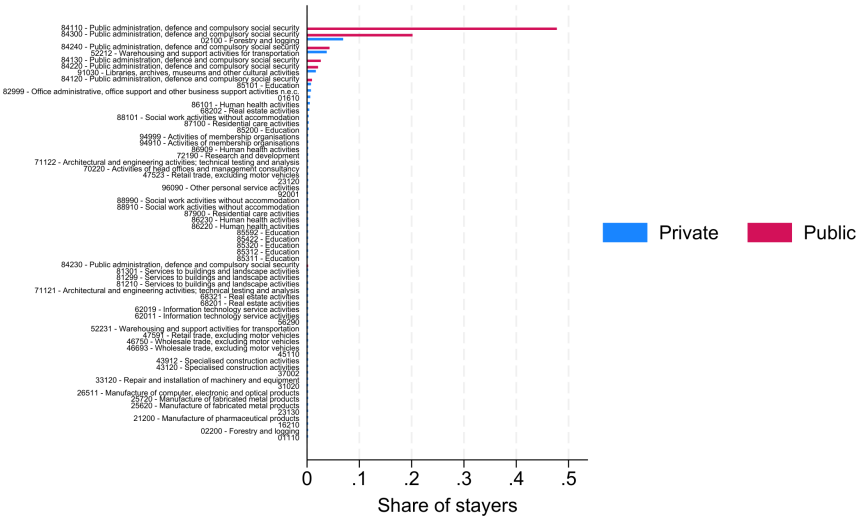
I distinguish *movers* from *commuters* among incumbents employed at receiving establishments in 2024. Movers have changed their recorded municipality of residence since 2014. Commuters still live in the same municipality. Movers account for 1.8% of the original workforce, while commuters account for 10.3% (Table 3). Among incumbents employed at receiving establishments, 85.2% have kept their original residence. Most incumbents who take jobs at the destinations therefore do so without moving home.

Most incumbents do not work at receiving establishments in 2024. I classify them as *stayers* if they retain their recorded municipality of residence and remain in regular employment outside the receiving establishments. Stayers account for 40.7% of the original workforce. Another 40.0% are no longer observed in regular social-security employment in 2024 and are very likely retirees. The remaining 7.2% have other or unclassified outcomes.

Stayers usually continue working in the public sector. Among stayers with a recorded employment sector in 2024, 78.0% work in public administration, defence, or compulsory social security (Figure 2). Within this same sample, general public administration accounts for 47.8% and compulsory social security for 20.2%. The remaining 22.0% work in other sectors, including forestry, transport support, cultural activities, and education.



(a) Broad industries



(b) Detailed activities

Figure 2. Employment sectors of stayers in 2024. Shares among original sending workers who retain their municipality of residence, work outside receiving establishments, and have a recorded sector in 2024.

Table 3. Worker adjustment and recruitment shares

Worker outcome	Share (%)
Original sending workforce, observed in 2014	
Receiving employment, changed residence	1.8
Receiving employment, unchanged residence	10.3
Employment elsewhere, unchanged residence	40.7
No longer observed in regular employment	40.0
Other or unclassified outcomes	7.2
Receiving workforce, observed in 2024	
Workers from outside the original sending workforce	99.1
Workers from the original sending workforce	0.9

Notes: The first panel reports 2024 outcomes as shares of the 2014 workforce at closed or substantially contracting sending establishments. Residence refers to the recorded municipality of residence. Receiving employment refers to employment at opened or substantially expanding receiving establishments. The second panel reports shares of the 2024 receiving workforce.

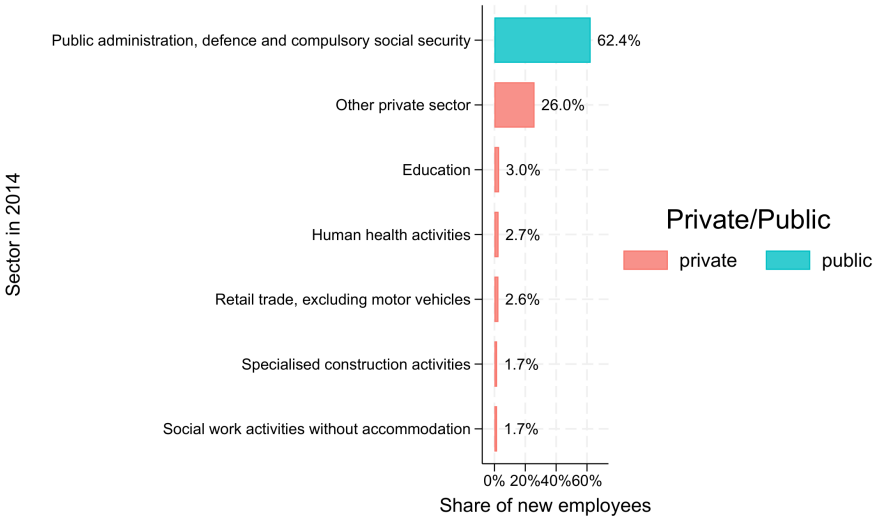
4.2 The Workforce at Receiving Agencies

Receiving agencies fill their positions overwhelmingly with workers from outside the original sending workforce. I call these workers *new hires*. They account for 99.1% of employment at receiving establishments in 2024, while incumbents account for 0.9% (Table 3). The term identifies workers who were absent from the original sending workforce. It does not require them to be entering employment for the first time.

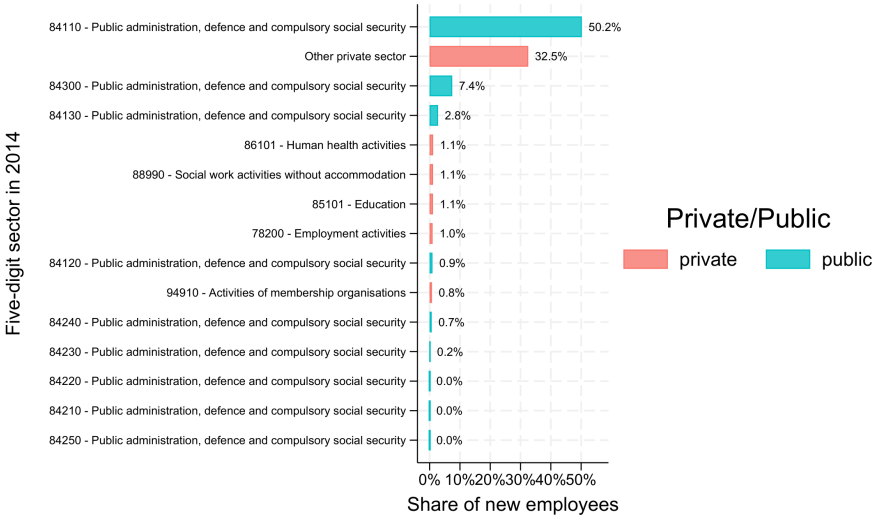
Most new hires with a recorded employment sector in 2014 already worked in the public sector before the program. Among these workers, 62.4% previously worked in public administration, defence, or compulsory social security, and 37.6% worked in other sectors (Figure 3). General public administration supplied 50.2%, compulsory social security 7.4%, and the administration of economic activities 2.8%. Other recruits came from education, health, retail, construction, and social work. Receiving agencies thus recruit heavily from workers with prior public sector employment, although almost none belonged to the original sending workforce.

To compare workforce characteristics, the agency-profile figures focus on establishments that close completely at the origins and establishments that open at the destinations. They compare the closed establishments' workers in 2014 with the newly opened establishments' workers in 2024. The respective municipalities' 2014 averages provide a local benchmark.

Women make up a larger share of employment at newly opened receiving agencies. Their share is 64.1% in 2024, compared with 48.5% at closed sending agencies in 2014, a difference of 15.7 percentage points (Figure 4). It also exceeds the receiving municipalities' 2014 average of 47.7% by 16.4 percentage points. The broader worker groups show the same pattern. Women account for 63.7% of new hires and 52.7% of stayers, a difference of 11.0 percentage points. Recruitment at the destinations therefore gives receiving agencies a substantially more female workforce than the former sending agencies.

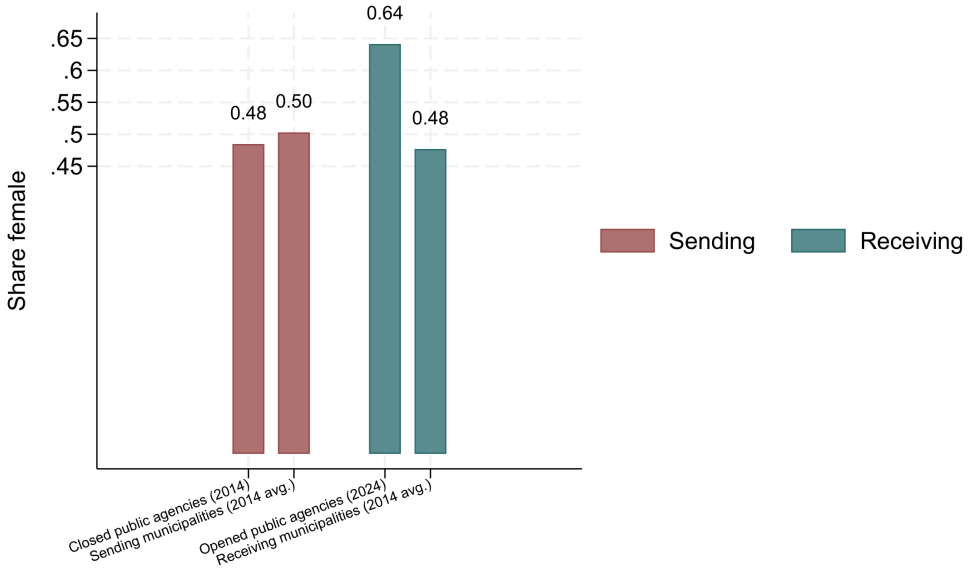


(a) Broad industries



(b) Detailed activities

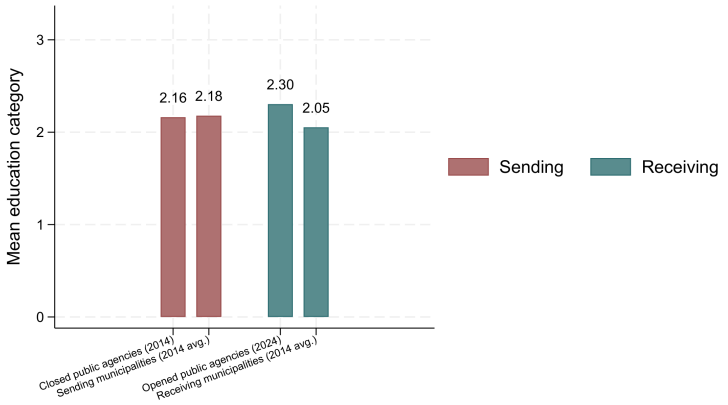
Figure 3. New hires' employment sectors before the program. Shares among workers at receiving establishments in 2024 who were outside the original sending workforce and have a recorded sector in 2014.



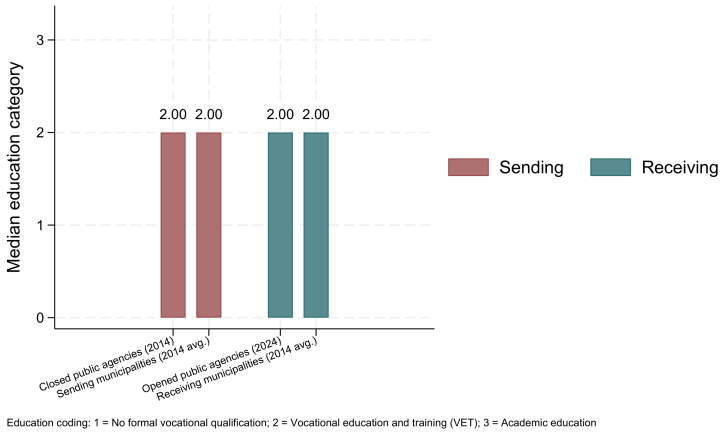
(a) Female employment shares

Figure 4. Female employment shares at sending and receiving agencies. Closed sending agencies in 2014 and newly opened receiving agencies in 2024 are compared with their respective municipalities' 2014 averages.

Workers at newly opened agencies tend more toward tertiary education than workers at closed sending agencies or in the receiving municipalities' 2014 workforce (Figure 5). The education measure distinguishes three levels of qualification, namely no formal vocational qualification, vocational training, and tertiary education such as a university degree. The worker-group comparison points in the same direction. New hires have higher average educational attainment than stayers.



(a) Mean educational attainment



(b) Median educational attainment

Figure 5. Educational attainment at sending and receiving agencies. The panels compare closed sending agencies in 2014 and newly opened receiving agencies in 2024 with their municipalities' 2014 averages. Categories distinguish no formal vocational qualification, vocational training, and tertiary education.

4.3 Wages

Workers at newly opened receiving agencies earn less on average than workers at closed sending agencies. All wage comparisons in this section are nominal. The receiving agencies' mean daily wage in 2024 is 4.8% below the closed sending agencies' mean in 2014. The median wage is also 4.8% lower. Yet the receiving agencies' mean wage exceeds the receiving municipalities' 2014 average by 10.8%. The agencies thus provide comparatively well-paid employment at the destinations, while employing workers who earn less than the former sending agencies' workforce.

Average nominal wages rise between 2014 and 2024 for both stayers and new hires. Stayers' mean daily wage rises by 8.3%. For workers classified as new hires in 2024, the group's mean wage is 10.1% above its 2014 mean. Despite this larger increase, new hires earn 3.3% less on average than stayers in 2024.

The location of public jobs and the residence of public employees adjust differently. Receiving agencies recruit mainly from outside the original workforce, while many incumbents retain their residence and continue working in the public sector. Receiving municipalities can therefore gain public jobs without attracting the original employees as residents. This distinction motivates the model's treatment of public workplaces as assigned by the government and residences as chosen by workers. The municipal analysis examines how private employment, commuting, and factor prices respond where public jobs arrive and where they leave.

5 Empirical Methodology

I estimate how local labor markets adjust when public jobs arrive and when they leave. For each receiving municipality, I construct a comparison with a similar untreated municipality; I repeat the procedure separately for sending locations. Event studies trace adjustment around the first relocation.

5.1 Matching and comparison groups

The policy directs public jobs toward places classified as disadvantaged relative to the state average. I reproduce the eligibility indicators for municipalities outside the policy state, identifying places that would qualify under the same benchmark. I match receiving municipalities to these comparison places and sending municipalities to similarly thriving locations. The matching accounts for the economic and demographic dimensions of the policy rule, municipal size, and pre-policy outcome trajectories. Within each comparison pool, I select a nearest neighbor by Mahalanobis distance, without replacement, separately for receiving and sending municipalities.

I choose comparison municipalities outside Bavaria to limit contamination by policy spillovers within the state. Even municipalities that neither receive nor send an agency may be affected as workers, firms, and expenditure adjust across nearby locations. As an additional precaution, I exclude municipalities in the 20 non-Bavarian counties with the

strongest commuting links to Bavaria.[11]

A causal interpretation requires that, without the relocation, treated municipalities would have followed the same outcome trends as their matched controls. Matching on pre-policy levels and changes addresses observed differences relevant to this comparison. The design also requires that anticipation does not change outcomes during the comparison's pre-treatment period. Agencies may prepare their premises and employees may adjust before operations begin, so I examine pre-treatment coefficients.

5.2 Staggered Difference-in-Differences

I use the estimator of Sun and Abraham (2021) as the main staggered difference-in-differences specification. It traces employment before and after treatment while allowing the response to differ across cohorts of municipalities first treated in different years. I estimate receiving and sending effects separately, using their matched never-treated municipalities as the comparison groups.

Treatment begins in the year of the first recorded relocation into a receiving municipality or out of a sending municipality. I use the official treatment year where it is available and the recorded completion year otherwise. Some municipalities experience further relocations, so first treatment marks the start of exposure rather than the completion of every project. Figure 6 shows the timing of individual projects by municipality and program; Table 14 reports first- and last-treatment cohorts in the matching roster. The distinction is especially relevant for places receiving several agencies over time.

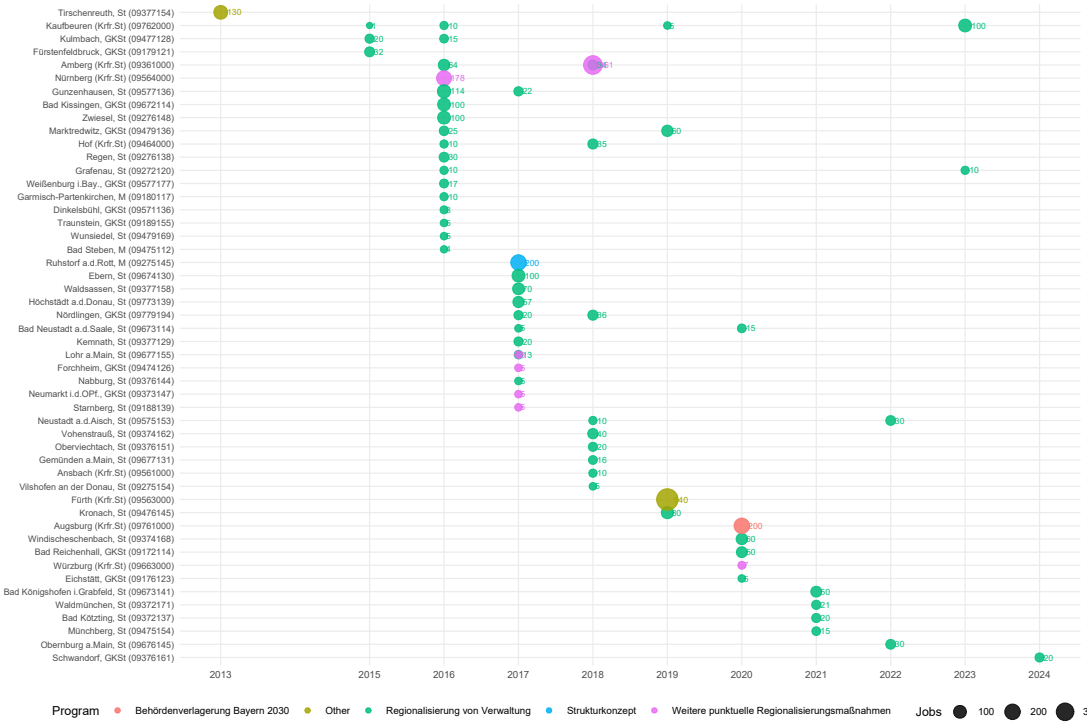
Let G_i denote municipality i's first treatment year. I estimate cohort-specific event-time effects using

$$Y_{it} = \alpha_i + \delta_{p(i),t} + \sum_{g \in \mathcal{G}} \sum_{\ell \neq -1} \theta_{g\ell} \, \mathbf{1}\{G_i = g\} \, \mathbf{1}\{t - g = \ell\} + u_{it}, \tag{1}$$

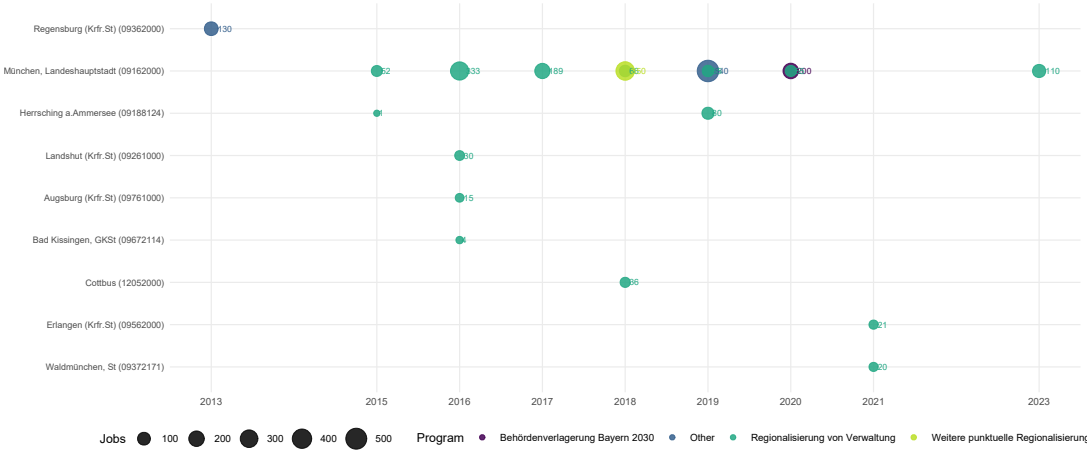
where α_i absorbs time-invariant municipal characteristics and $\delta_{p(i),t}$ absorbs shocks common to a matched pair in a given year. The year before first treatment is the reference period. I aggregate the cohort-specific coefficients at each event time to obtain the reported event-study effects. The main employment-count event studies use levels, so their coefficients measure employees. Confidence intervals allow for clustering by municipality.

I also estimate the group-time treatment effects of Callaway and Sant'Anna (2021) as a complementary specification. These estimates use never-treated controls and a common pre-treatment reference year within each cohort, with bootstrap confidence bands for the dynamic effects. Appendix C documents the aggregation and inference. The next section presents the municipal responses at both ends of the relocations.

[11] The exclusion uses the counties identified by their shares of commuters to Bavaria in 2013. Appendix C shows their geography and documents the matching procedure.



(a) Receiving locations



(b) Sending locations

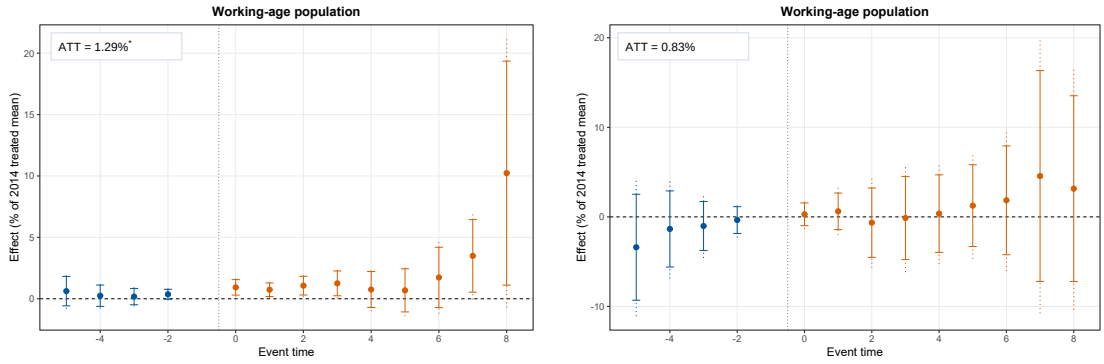
Figure 6. Timing of relocation projects by municipality and program. The figures show the broader program inventory; the matched-sample first- and last-treatment cohorts are reported in Table 14.

6 Place-Level Effects

Receiving municipalities gain private sector jobs as public employment expands, while sending municipalities show no evidence of offsetting private employment losses. Five years after the first relocation, the receiving private employment gain amounts to 4.3% relative to matched controls. I trace how population, employment, commuting, and factor prices adjust at both ends of the program. The figures report Sun–Abraham event studies around each municipality's first relocation, comparing treated municipalities with matched controls outside the policy state. I express employment and population responses as percentages compared to 2014 treated means for each outcome. The appendix reports additional outcomes.

6.1 Population Adjustment

Receiving municipalities gain working-age residents, although the population response is modest. Their working-age population rises by 0.9% in the year of the first relocation and remains about 1% higher than controls over the subsequent three years (Figure 7). The average post-treatment gain is 1.3%. In sending municipalities, working-age population rises by 0.8%. However, the population response in the sending locations is not statistically significant.



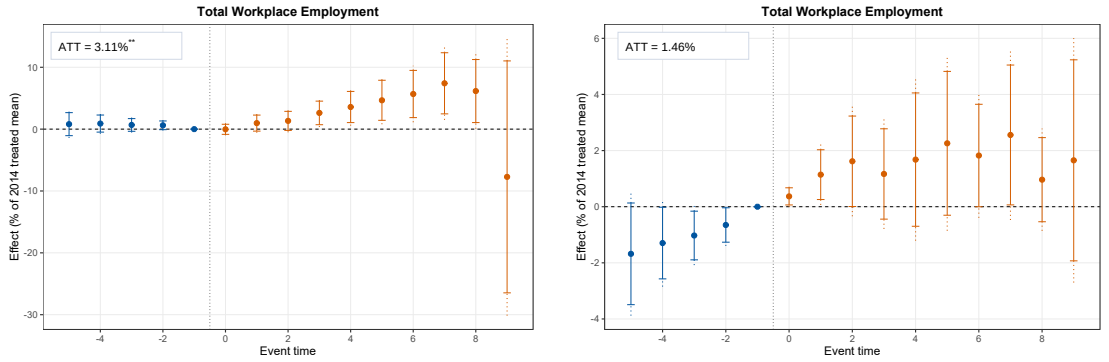
(a) Receiving. The 2014 treated mean is 9,831 residents. **(b)** Sending. The 2014 treated mean is 286,283 residents.

Figure 7. Working-age population around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Residents of working age. The series begin in 2011. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6.2 Employment Outcomes

Receiving municipalities gain workplace employment gradually after the first agency arrives. Total employment, including public and private jobs, rises by 4.7% after five years and 7.4% after seven years (Figure 8). The average post-treatment gain is 3.1%. Sending

municipalities also have positive employment estimates throughout the post-treatment period, averaging 1.5%.



(a) Receiving. The 2014 treated mean is 8,857 employees.

(b) Sending. The 2014 treated mean is 228,704 employees.

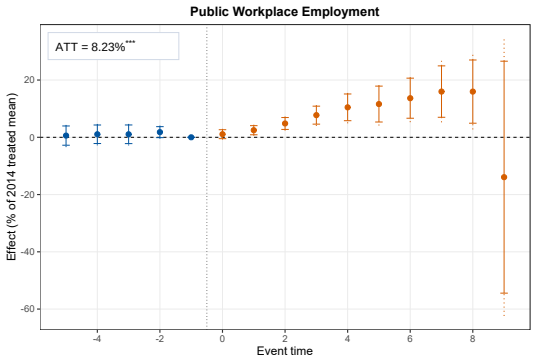
Figure 8. Total workplace employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public and private employment. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

I next separate the total employment response into its public and private components. Public employment builds up as receiving municipalities gain relocated agencies and those agencies fill their positions. It rises by 4.8% two years after the first relocation and by 13.7% after six years (Figure 9). The average post-treatment increase is 8.2%. This gradual expansion fits the phased implementation described in Section 2. In sending municipalities, municipal public employment does not fall immediately. The estimates turn negative six years after the first relocation and reach a decline of 3.1% after seven years. The trajectories suggest that the receiving public sector expands before a contraction appears in municipal employment at the origins.

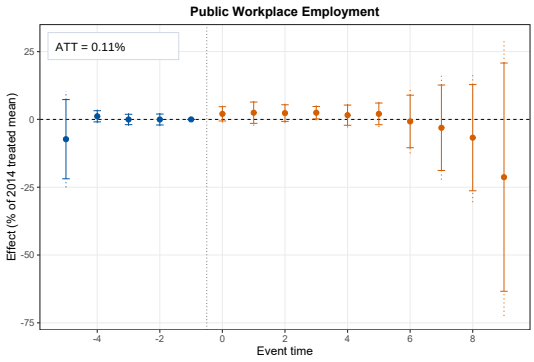
Private employment expands alongside public employment in receiving municipalities. The response is small during the first two years, then grows to 2.3% after three years, 3.2% after four years, and 4.3% after five years (Figure 10). Averaged across the post-treatment period, the increase is 2.8%. Receiving municipalities thus gain jobs beyond those located in the public agencies themselves, with the private response taking three years to develop.

Sending municipalities' private employment estimates point to expansion rather than to losses that offset the gains at the destinations. They are positive in every post-treatment year, averaging 1.5% and reaching 2.3% after five years (Figure 10).

Receiving municipalities draw more workers from surrounding places as their employment grows. In-commuting rises by 2.4% one year after the first relocation and by 5.9% after five years (Figure 11). The average post-treatment increase is 3.8%. The additional jobs therefore reach workers who live outside the receiving municipality, consistent with the distinction between jobs and residence in the worker evidence. Sending municipalities also have a positive average in-commuting response of 9.2%, suggesting that they continue

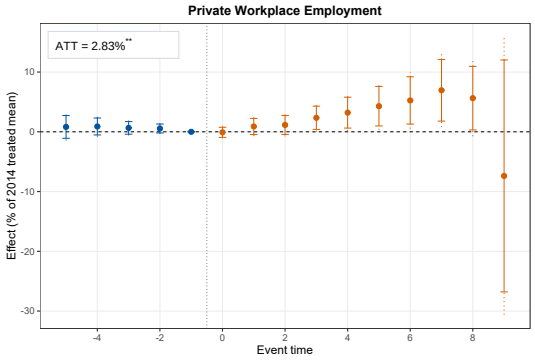


(a) Receiving. The 2014 treated mean is 450 employees.

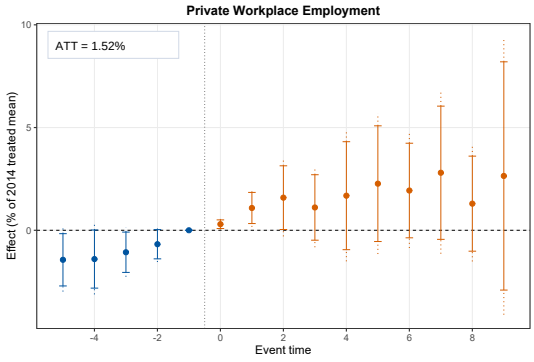


(b) Sending. The 2014 treated mean is 9,474 employees.

Figure 9. Public workplace employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in public administration, defense, and compulsory social security. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



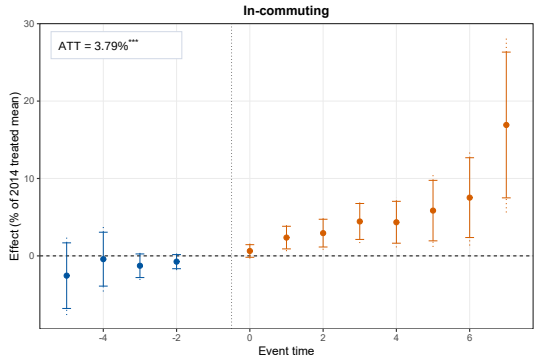
(a) Receiving. The 2014 treated mean is 8,407 employees.



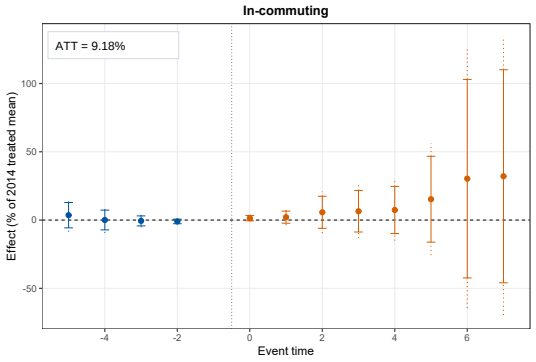
(b) Sending. The 2014 treated mean is 219,230 employees.

Figure 10. Private workplace employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Employees in industries outside public administration. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

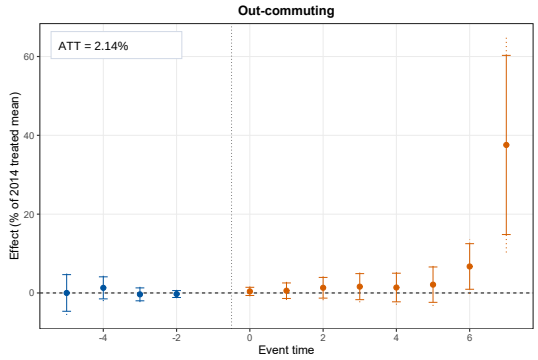
to attract workers from surrounding places. However, the results are not statistically significant.



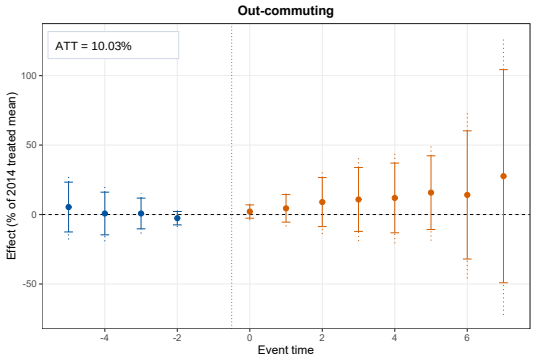
(a) In-commuting: Receiving. The 2014 treated mean is 5,172 commuters.



(b) In-commuting: Sending. The 2014 treated mean is 115,581 commuters.



(c) Out-commuting: Receiving. The 2014 treated mean is 2,891 commuters.



(d) Out-commuting: Sending. The 2014 treated mean is 47,419 commuters.

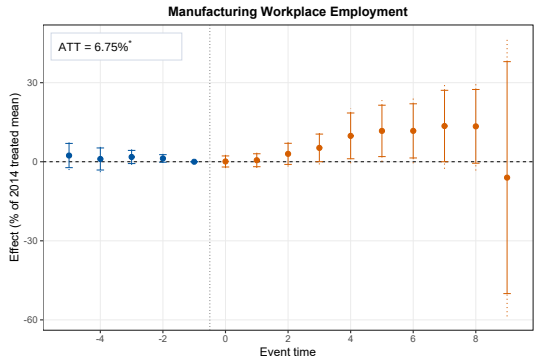
Figure 11. Commuting around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Commuters across municipal boundaries. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Out-commuting initially remains unchanged in both receiving and sending municipalities (Figure 11). In receiving municipalities, it begins to rise six to seven years after the first relocation, reaching 6.7% after six years and 37.6% after seven years. One possible interpretation is that people who move with agency employees commute to jobs outside the receiving municipality. Further study would be needed to establish whether such co-movers explain the increase.

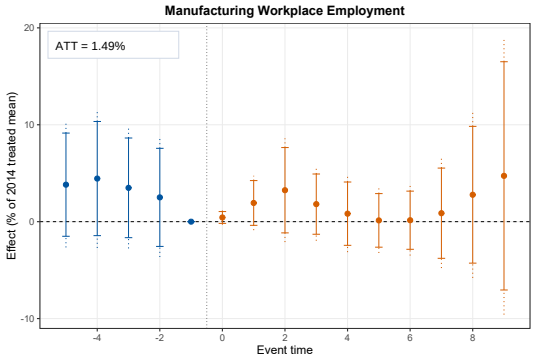
6.3 Sector Employment

I next examine how these employment responses differ across sectors. Manufacturing accounts for much of the employment expansion in receiving municipalities. Its response

develops with a similar delay to the private employment response: manufacturing employment rises by 5.3% after three years and by 11.7% after five years (Figure 12). The average post-treatment gain is 6.7%. The expansion therefore extends to a sector that serves markets beyond the municipality pointing to a spillovers channel beyond local consumption. Sending municipalities have a smaller positive, but not statistically significant average manufacturing response of 1.5% that seems cyclical and unrelated to the policy implementation.



(a) Receiving. The 2014 treated mean is 2,292 employees.



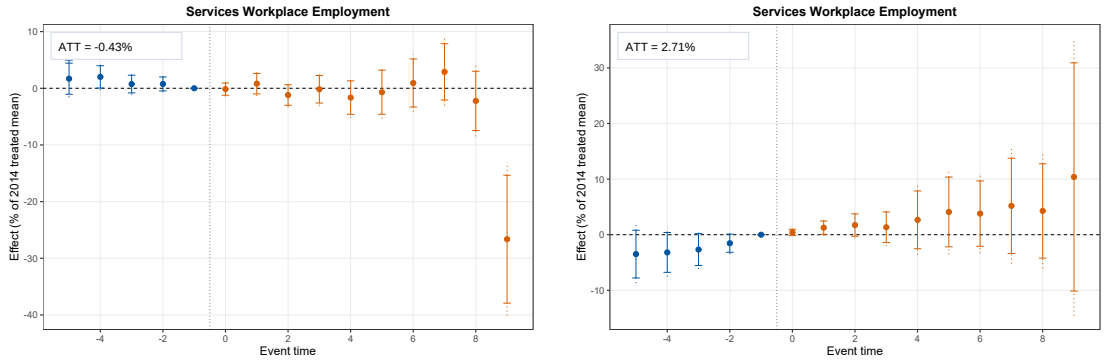
(b) Sending. The 2014 treated mean is 31,217 employees.

Figure 12. Manufacturing employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Aggregate service employment changes little in receiving municipalities and rises modestly in sending municipalities. The average responses are −0.4% and 2.7%, respectively (Figure 13). Neither aggregate change is statistically significant. Breaking down services by subsector reveals different sources of employment growth at the two ends of the program: knowledge services expand in sending municipalities, while food and beverage services expand in receiving municipalities. Arts, entertainment, and recreation remain broadly unchanged in both groups.

Knowledge services expand in sending municipalities as public jobs move elsewhere. Knowledge-intensive services have an average post-treatment increase of 5.2%, while the narrower knowledge-intensive business services category increases by 12.2% (Figure 14). The larger proportional response in business services points to growth in activities that serve other firms. Knowledge services therefore explain most of the expansion of private employment in sending municipalities.

Food and beverage services expand in receiving municipalities, consistent with increased demand for local consumption. Employment in these services rises by 5.4% in the year of the first relocation and remains about 7% higher over the next two years (Figure 15). The average post-treatment increase is 7.8%. This early response precedes the larger manufacturing gains and suggests that local consumption contributes to the



(a) Receiving. The 2014 treated mean is 3,651 employees.

(b) Sending. The 2014 treated mean is 142,306 employees.

Figure 13. Service employment around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

receiving municipalities' private employment growth from the start of the program. Arts, entertainment, and recreation remain broadly unchanged in both receiving and sending municipalities, so the expansion of local consumption services is concentrated in food and beverage activities.

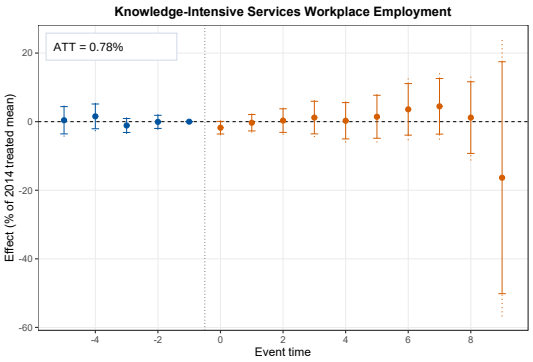
6.4 Employment by Gender

The relocations change the gender composition of public and private employment in opposite directions. The worker-level results in Section 4 show that women make up a larger share of employment in receiving agencies after the move. In the private sector of receiving municipalities, however, the female employment share decreases. The agencies become more female, while the composition of private employment shifts toward men. These patterns suggest that gender differences in sector specialization relocate with the agencies.

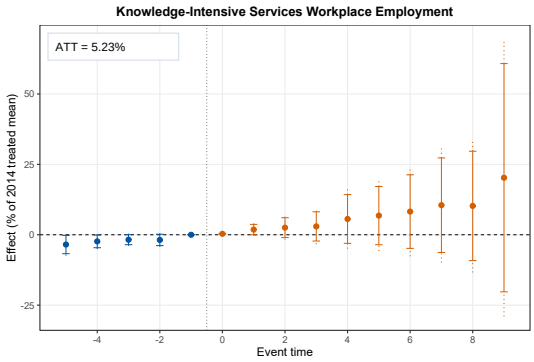
6.5 Factor Prices

Private wages rise in receiving municipalities and decrease slightly in sending municipalities. The average post-treatment changes are approximately 0.4% and −1.1%, respectively (Figure 17). Receiving municipalities therefore gain private employment alongside a modest rise in average private pay, while private employment growth in sending municipalities is accompanied by slightly lower average pay. Neither *private* wage trajectory is statistically significant.

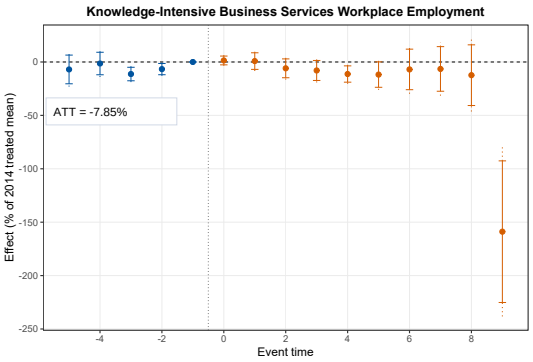
Public wages move in the opposite direction: average public workplace wages decrease in receiving municipalities and increase in sending municipalities. The average post-treatment changes are approximately −2.6% and 5.0%, respectively (Figure 18). Changes



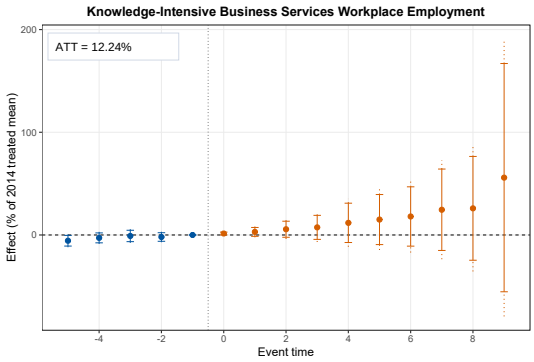
(a) Knowledge-intensive services: Receiving. The 2014 treated mean is 2,549 employees.



(b) Knowledge-intensive services: Sending. The 2014 treated mean is 93,378 employees.

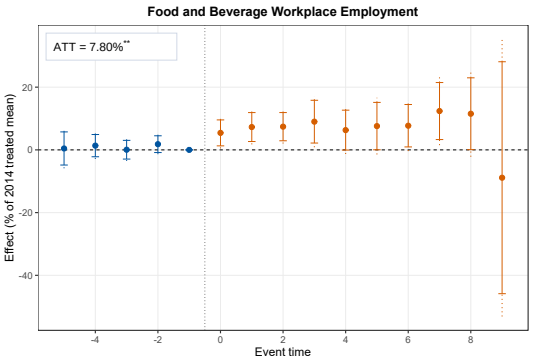


(c) Knowledge-intensive business services: Receiving. The 2014 treated mean is 394 employees.

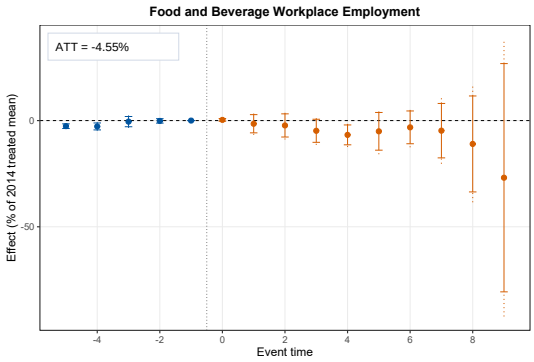


(d) Knowledge-intensive business services: Sending. The 2014 treated mean is 37,506 employees.

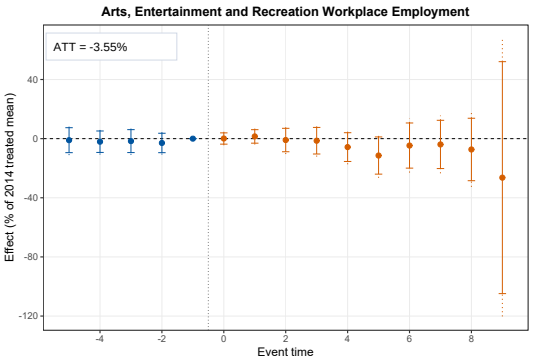
Figure 14. Knowledge services around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



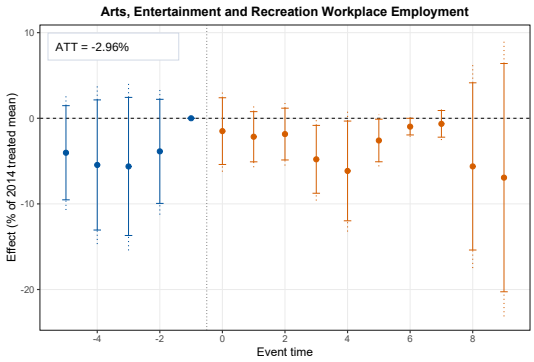
(a) Food and beverage: Receiving. The 2014 treated mean is 323 employees.



(b) Food and beverage: Sending. The 2014 treated mean is 9,703 employees.

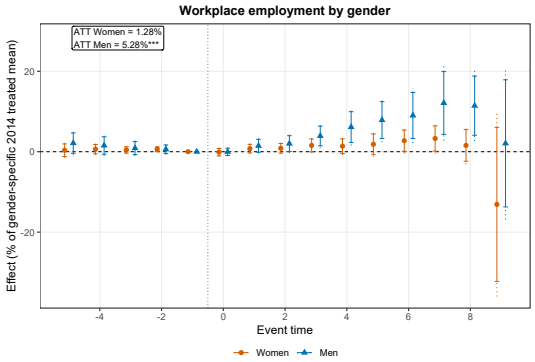


(c) Arts, entertainment, and recreation: Receiving. The 2014 treated mean is 91 employees.

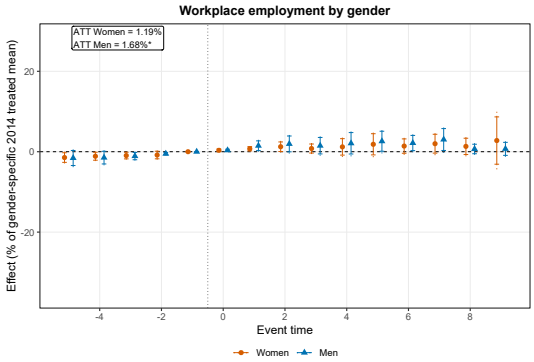


(d) Arts, entertainment, and recreation: Sending. The 2014 treated mean is 3,108 employees.

Figure 15. Local consumption services around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Workplace employees. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

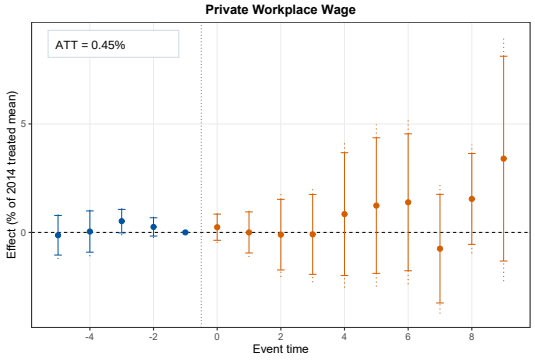


(a) Receiving. The 2014 treated means are 4,699 women and 4,230 men.

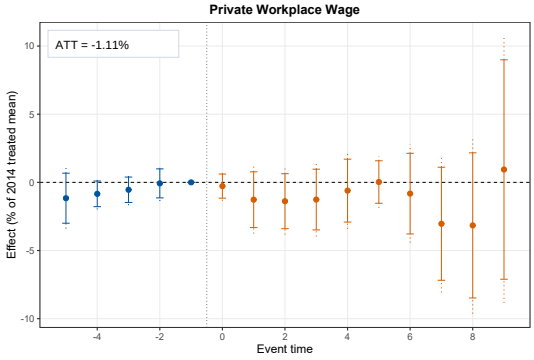


(b) Sending. The 2014 treated means are 111,689 women and 117,922 men.

Figure 16. Workplace employment by gender around the first relocation. Sun–Abraham estimates relative to matched municipalities outside the policy state. The year before treatment is the reference. Estimates and confidence intervals are expressed as percentages of the respective gender's 2014 treated mean. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



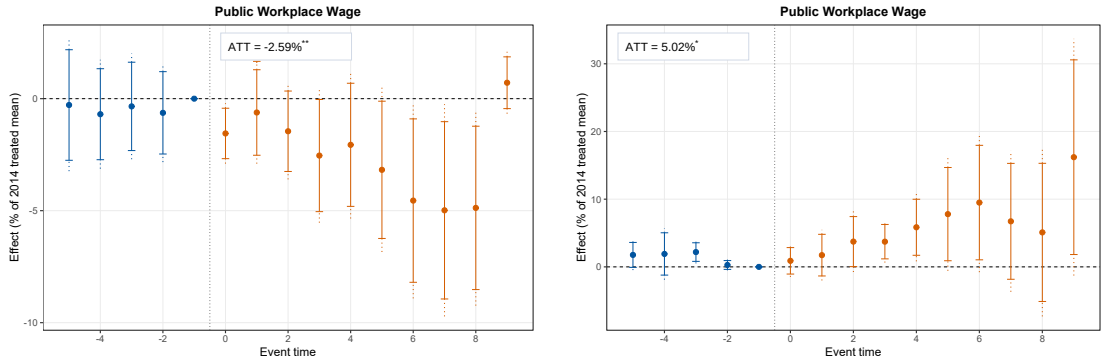
(a) Receiving. The 2014 treated mean is 26 thousand euros.



(b) Sending. The 2014 treated mean is 30 thousand euros.

Figure 17. Private workplace wages around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Nominal gross wages. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

in the workforce offer a possible explanation. The receiving agencies hire younger workers who earn less than the former agency workforce, while many incumbent workers remain in the sending locations at other public agencies where their nominal wages rise. The municipal wage pattern is consistent with those descriptive findings. Average public wages can therefore change as the composition of employment changes, without every incumbent worker experiencing the same wage change. Both average public wage responses are statistically significant.



(a) Receiving. The 2014 treated mean is 32 thousand euros.

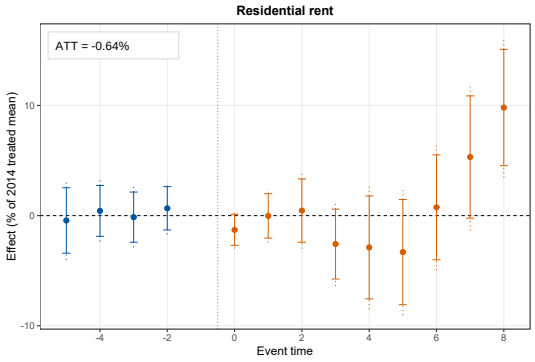
(b) Sending. The 2014 treated mean is 35 thousand euros.

Figure 18. Public workplace wages around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Nominal gross wages. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

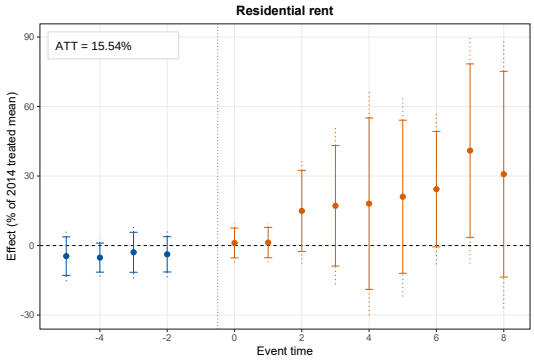
Residential rents in receiving municipalities initially fall modestly and rise at longer horizons, despite the earlier growth in local employment. Five years after the first relocation, the estimate is -3.3%;++, and reaches 9.8% after eight years (Figure 19). The average post-treatment response is a small decline of -0.6%. The trajectory suggests that employment expands before higher rents emerge in receiving municipalities. Sending municipalities have an average residential rent increase of 15.5%, pointing to a substantial rise in housing costs as public jobs move elsewhere and private jobs increase.

Commercial rents rise modestly on average in receiving municipalities. The average post-treatment response is 2.4%, with a larger increase of 7.7% five years after the first relocation (Figure 20). The pattern is consistent with additional demand for office space from public agencies and private activity. Sending municipalities have a larger positive average response of 10.0%, pointing to rising office rents despite the release of public office space. Freeing public office space need not reduce rents if private firms take up that space and expand. The model in Section 7 accounts for both the release of floor space and the response of private firms.

The municipal employment gains leave open how far the benefits extend and what they imply for the economy as a whole. More in-commuting means that some of the workers filling jobs in receiving municipalities live elsewhere, and local employment growth may

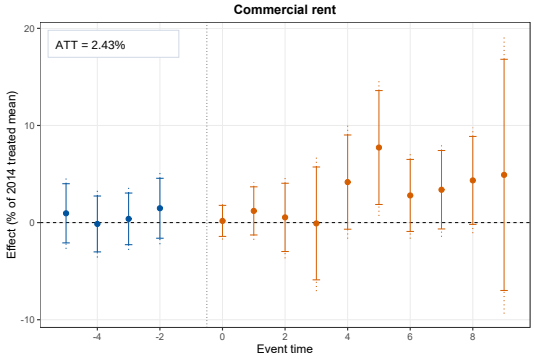


(a) Receiving. The 2014 treated mean is 5.59 euros per square meter.

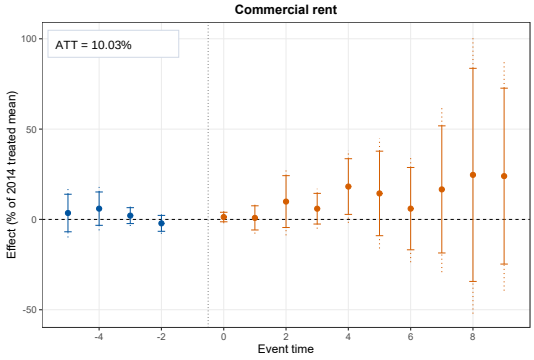


(b) Sending. The 2014 treated mean is 10.40 euros per square meter.

Figure 19. Residential rents around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Receiving. The 2014 treated mean is 6.71 euros per square meter.



(b) Sending. The 2014 treated mean is 9.36 euros per square meter.

Figure 20. Commercial rents around the first relocation. Sun–Abraham estimates around the first relocation, relative to matched municipalities outside the policy state. The year before treatment is the reference. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

draw activity from nearby places. The quantitative spatial model accounts for adjustments across locations and evaluates how public-to-private agglomeration economies and the use of floor space translate local employment responses into aggregate output and welfare.

7 Two-Sector Quantitative Spatial Model

Receiving municipalities gain private employment even though few incumbent public employees move their residence with the agencies. These local gains raise two questions: how do public workplaces support private activity, and what do the relocations imply for the economy as a whole? I use a quantitative spatial model to connect the worker and place-level evidence to private production, its distribution across Germany, and private workers' living standards. The model separates the location of public jobs from the residence choices of their holders, and it allows places outside the program to respond.

Two mechanisms connect public employment to private activity. Public workplaces generate productivity spillovers for nearby private firms, distinct from the spillovers that private firms generate for one another. Public agencies also occupy floor space. Moving an agency releases space at its origin and creates demand for space at its destination. Private firms and households compete for the space left after public offices. A common labor-income tax finances public payroll and office costs, so changes in these costs also affect disposable income.

7.1 Model Setup

I extend the spatial framework of Ahlfeldt, Redding, et al. (2015) to include public workers with assigned workplaces. Locations cover all of Germany, with i indexing residences and j workplaces. Each location has land area $K_i > 0$ and effective floor-space stock $L_i > 0$. I implement the model at the county level. The national masses of private and public workers, $\bar{H}^P$ and $\bar{H}^G$, are fixed. Each worker supplies one unit of labor, and sector membership is predetermined. The equilibrium reallocates employed workers across locations.

The government assigns $\bar{H}^G_{Mj}$ public jobs to workplace j, with $\sum_j \bar{H}^G_{Mj} = \bar{H}^G$. Public workers choose where to live conditional on that assignment, making public resident employment, H^G_{Ri}, endogenous. Private workers choose both residence and workplace, generating private workplace employment H^P_{Mj} and private resident employment H^P_{Ri}. Total workplace and resident employment are

$$H_{Mj} = H^P_{Mj} + \bar{H}^G_{Mj}, \qquad H_{Ri} = H^P_{Ri} + H^G_{Ri}.$$

Resident employment refers to employed people living in a location, wherever they work. Both sectors receive the same endogenous gross nominal wage at workplace j, $w^P_j = w^G_j = w_j$, determined by private labor demand. A relocated public job receives the wage at its destination. All wages in this chapter are nominal wages.

7.1.1 Workers

Workers value consumption, housing, amenities, and access to work. They spend share β of disposable labor income on a freely traded numeraire good and share $1-\beta$ on residential space. Let Q_i^R denote residential rent, t the common proportional labor-income tax, and z_{ijo}^m worker o's preference draw for an alternative in sector $m \in \{P, G\}$. Indirect utility is

$$u_{ijo}^m = \frac{z_{ijo}^m B_i (1-t) w_j}{(Q_i^R)^{1-\beta} d_{ij}}, \qquad d_{ij} = \exp(\kappa \tau_{ij}), \tag{2}$$

where B_i is the amenity level at residence i, τ_{ij} is road travel time in minutes, and κ governs commuting costs. Commuting reduces utility without subtracting cash from the household budget. Independent Fréchet preference draws have common shape $\epsilon > 1$ and scales $T_i^P E_j^P$ for private residence–workplace pairs and T_i^G for public residences conditional on assignment. These scales distinguish tastes from physical amenities and wages.

Private workers compare all residence–workplace pairs. Define $D_{ij} = \exp(-\epsilon\kappa\tau_{ij})$ and $Z_i^m = T_i^m B_i^\epsilon (Q_i^R)^{-\epsilon(1-\beta)}$. Private choice probabilities and bilateral worker flows are

$$\phi_{ij}^P = Z_i^P D_{ij} E_j^P [(1-t)w_j]^\epsilon, \quad \Phi^P = \sum_{r,s} \phi_{rs}^P, \quad \pi_{ij}^P = \frac{\phi_{ij}^P}{\Phi^P}, \quad N_{ij}^P = \bar{H}^P \pi_{ij}^P. \tag{3}$$

Summing N_{ij}^P across residences gives workplace employment, $H_{Mj}^P = \sum_i N_{ij}^P$; summing across workplaces gives resident employment, $H_{Ri}^P = \sum_j N_{ij}^P$. Higher wages make workplaces more attractive. Higher housing costs and longer commutes make the corresponding residence–workplace pairs less attractive.

Public workers compare residences for their assigned workplace. Writing $\Psi_j^G = \sum_r Z_r^G D_{rj}$, their conditional probabilities and flows are

$$\pi_{i|j}^G = \frac{Z_i^G D_{ij}}{\Psi_j^G}, \qquad N_{ij}^G = \bar{H}_{Mj}^G \pi_{i|j}^G, \qquad H_{Ri}^G = \sum_j N_{ij}^G. \tag{4}$$

The workplace wage and common tax cancel from this residential comparison because they are the same for every residence available to the assigned job. They still determine income, housing demand, and utility. Moving a public workplace changes its commuting access without mechanically moving its workers' residences. This distinction carries the worker evidence into the model.

The choice sets determine expected utility. With $\Gamma_\epsilon = \Gamma(1 - 1/\epsilon)$,

$$\bar{U}^P = \Gamma_\epsilon (\Phi^P)^{1/\epsilon}, \qquad \bar{U}^{G|j} = \Gamma_\epsilon (1-t) w_j (\Psi_j^G)^{1/\epsilon}. \tag{5}$$

Private expected utility values access to the full set of private residence–workplace pairs. Public expected utility is conditional on an assigned workplace. Both incorporate disposable wages, housing costs, amenities, and commuting. Appendix 8 derives the choice probabilities and employment margins.

7.1.2 Private Firms

Private firms produce the freely traded good using labor and private commercial floor space L_j^P:

$$Y_j^P = A_j^P (H_{Mj}^P)^{\alpha} (L_j^P)^{1-\alpha}. \tag{6}$$

Firms take productivity A_j^P, wages, and commercial rent q_j as given. Competition and constant returns imply

$$w_j H_{Mj}^P = \alpha Y_j^P, \qquad q_j L_j^P = (1-\alpha) Y_j^P. \tag{7}$$

Public offices are separate from L_j^P. Vacated offices can therefore increase the space available for private production or housing, with rents and demand determining its use.

7.1.3 Within- and Between-Sector Productivity Spillovers

Private productivity depends separately on nearby private and public workplace density:

$$A_j^P = a_j (\Upsilon_j^P)^{\lambda_{PP}} (\Upsilon_j^G)^{\lambda_{PG}}, \tag{8}$$

$$\Upsilon_j^m = \sum_s \exp(-\delta_A \tau_{js}) \frac{H_{Ms}^m}{K_s}, \qquad m \in \{P, G\}, \quad H_{Ms}^G = \bar{H}_{Ms}^G. \tag{9}$$

The productivity fundamental a_j is fixed across policy scenarios. The elasticities λ_{PP} and λ_{PG} govern private-to-private and public-to-private spillovers, while δ_A governs their decay with travel time. Each source contributes employment per unit of its own land area; the travel-time weights are not normalized. Moving public jobs changes the public density term directly. The subsequent response of private firms and workers changes the private density term. I maintain the private elasticity from Ahlfeldt, Redding, et al. (2015) and estimate the public elasticity from the relocation responses.

7.1.4 Amenities, Floor Space, and Market Clearing

Amenities depend on where workers live:

$$B_i = b_i \Omega_i^{\eta}, \qquad \Omega_i = \sum_s \exp(-\delta_B \tau_{is}) \frac{H_{Rs}^P + H_{Rs}^G}{K_s}. \tag{10}$$

Both worker types contribute to surrounding resident density. The amenity fundamental b_i is fixed, η is the amenity elasticity, and δ_B governs spatial decay. A relocation can consequently change workplace productivity more than residential amenities when workers commute to the new jobs from their existing homes.

Public offices occupy $\bar{L}_i^G$, leaving private firms and households to share the residual stock:

$$L_i = L_i^P + \bar{L}_i^G + L_i^R, \qquad F_i = L_i - \bar{L}_i^G > 0. \tag{11}$$

Office requirements are assigned with public employment. At baseline, a public job uses fraction γ_G of the floor space used per private worker in the same location:

$$\ell_i^{G,0} = \gamma_G \frac{1-\alpha}{\alpha} \frac{w_i^0}{q_i^0}, \qquad \bar{L}_i^{G,0} = \ell_i^{G,0} \bar{H}_{Mi}^{G,0}. \tag{12}$$

I set $\gamma_G = 0.25$, using observed requirements in new state offices relative to the model-implied floor space per private worker in Munich. Office requirements remain fixed during the equilibrium adjustment. A transferred position carries its origin's baseline requirement to its destination.

Income earned at workplaces finances housing at residences. Gross resident income is $I_i = \sum_j w_j (N_{ij}^P + N_{ij}^G)$, and disposable income is $v_i = (1-t) I_i$. Housing demand satisfies $Q_i^R L_i^R = (1-\beta) v_i$. Every county is in mixed use, and a fixed wedge ξ_i relates commercial and residential rents through $q_i = \xi_i Q_i^R$. I use the observed rent ratio, including ratios below one. Floor-space market clearing gives

$$q_i = \frac{(1-\alpha) Y_i^P + \xi_i (1-\beta) v_i}{L_i - \bar{L}_i^G}, \qquad Q_i^R = \frac{q_i}{\xi_i}. \tag{13}$$

Public occupancy reduces the space available to private uses. Rents also respond to private production and resident income, so releasing public offices does not by itself determine the direction of the rent response.

The private share of the residual stock differs from the commercial share of the total stock:

$$\vartheta_i = \frac{L_i^P}{L_i - \bar{L}_i^G} = \frac{(1-\alpha) Y_i^P}{q_i (L_i - \bar{L}_i^G)}, \qquad \theta_i^M = \frac{L_i^P + \bar{L}_i^G}{L_i}.$$

The first measures how private firms and households divide the space left after public offices. The second includes public offices as commercial use. A relocation can change both shares by changing public occupancy and the demand for private space.

The government finances public payroll and office costs with a common tax on labor income. Define $W^P = \sum_j w_j H_{Mj}^P$, $W^G = \sum_j w_j \bar{H}_{Mj}^G$, and $C^G = \sum_j q_j \bar{L}_j^G$. The balanced budget implies

$$t(W^P + W^G) = W^G + C^G, \qquad t = \frac{W^G + C^G}{W^P + W^G}. \tag{14}$$

Public wages are taxed on the same basis as private wages. The office bill values both leased and state-owned premises at their commercial rental equivalent. Rents accrue to absentee landlords and are not returned to resident households. The budget covers the modeled administration sector. Its payroll and office bill change with equilibrium wages and rents even though national public employment is fixed.

An equilibrium makes worker choices, private factor demand, assigned public employment, floor-space use, spillovers, and the government budget mutually consistent. Private labor supplied to each workplace equals firms' demand; public assignments are filled; and each sector's workplace and residence totals equal its national mass. Floor-space use is feasible, and $0 \le t < 1$, which requires $C^G < W^P$. Expected utility adjusts endogenously with the allocation of the fixed national workforce.

7.2 Quantification

The baseline combines Census employment by sector at workplaces and residences in 2011, county residential and office rents in 2011, Federal Employment Agency commuting flows in 2013, and road travel times in minutes (Section 3). I harmonize the inputs to county boundaries in 2011. Census industry O, public administration, defense, and compulsory social security, defines public employment; the other industries define private employment.

I take the production and consumption shares, private productivity elasticity, amenity elasticity, and spatial decay rates from Ahlfeldt, Redding, et al. (2015). I estimate the commuting decay $\nu = \epsilon\kappa$ from a gravity equation for county commuting flows, using Poisson pseudo-maximum likelihood with origin and destination fixed effects and a control function for travel time. In a preliminary private-sector commuting inversion, I choose the Fréchet shape ϵ to match the cross-county variance of log nominal wages. This gives $\epsilon = 2.486$ and $\kappa = \nu/\epsilon = 0.0172$ per minute. The observed commercial-to-residential rent ratio has median $\xi_i = 1.145$ and ranges from 0.76 to 2.05. The tax rate that balances the baseline government budget is 7.54%. Table 4 reports the parameters.

Table 4. Model parameters

Symbol	Description	Value	Source
Estimated			
λ_{PG}	Public-to-private productivity elasticity	0.364	Indirect inference
ϵ	Fréchet shape	2.486	Variance of log county wages
$\nu = \epsilon\kappa$	Commuting decay per minute	0.0429	County commuting gravity
Calibrated or derived			
κ	Commuting cost per minute	0.0172	ν/ϵ
ξ_i	Commercial-to-residential rent ratio	1.145	Observed rent ratio; median
γ_G	Public office intensity relative to private	0.25	New state-office requirements
t^0	Baseline labor-income tax rate	0.0754	Balanced government budget
Maintained			
λ_{PP}	Private-to-private productivity elasticity	0.071	Ahlfeldt, Redding, et al. (2015)
δ_A	Productivity decay per minute	0.3617	Ahlfeldt, Redding, et al. (2015)
η	Amenity elasticity	0.1553	Ahlfeldt, Redding, et al. (2015)
δ_B	Amenity decay per minute	0.7595	Ahlfeldt, Redding, et al. (2015)
α	Labor share in private production	0.80	Ahlfeldt, Redding, et al. (2015)
β	Consumption expenditure share	0.75	Ahlfeldt, Redding, et al. (2015)

Given the parameters, I invert the model to recover productivity and effective amenity fundamentals that rationalize the baseline. For quantification, I set $E_j^P = T_i^P = T_i^G = 1$, so both worker types share the residential attractiveness index $Z_i^P = Z_i^G = Z_i$. The inversion jointly matches private workplace employment and total resident employment. The split into private and public residents follows from their respective choice rules. I scale the recovered nominal wages to the observed geometric mean of county wages. Wages and rents recover private productivity and factor demand. Public office requirements and housing demand then complete the effective floor-space stock. Appendix 8 gives the inversion equations.

I estimate λ_{PG} by indirect inference. For each candidate value, I recover the baseline fundamentals, solve the relocation counterfactual, and compute the model-implied log change in private workplace employment, $g_j(\lambda_{PG}) = \log H_{Mj}^{P,\text{cf}} - \log H_{Mj}^{P,0}$. I apply the matched-pair regression to these simulated outcomes in receiving and sending county samples $\mathcal{S}_a$:

$$g_j(\lambda_{PG}) = \mu_{p(j),a} + \beta_a^M(\lambda_{PG})\text{Treat}_{j,a} + e_{j,a}, \qquad j \in \mathcal{S}_a, \tag{15}$$

where $\mu_{p(j),a}$ is a matched-pair fixed effect. The targets $\widehat{\beta}_a^{RF}$ come from the corresponding county regressions for the 2014–2024 log change. Control counties receive their model-implied outcomes, so the comparison incorporates policy effects outside the treated counties. I minimize the weighted distance between the simulated and reduced-form coefficients:

$$\widehat{\lambda}_{PG} = \arg\min_{\lambda_{PG}} J(\lambda_{PG}), \qquad J(\lambda_{PG}) = \sum_{a\in\{\text{receiving,sending}\}} \left[\frac{\beta_a^M(\lambda_{PG}) - \widehat{\beta}_a^{RF}}{s_a^{RF}}\right]^2, \tag{16}$$

with fixed weighting scales s_a^{RF} and a search over $\lambda_{PG} \in [-0.1, 1]$.

The estimated elasticity, $\widehat{\lambda}_{PG} = 0.364$, implies that a 1% increase in travel-time-weighted public workplace density raises private productivity by about 0.36%, holding private density and fundamentals fixed. This gives public workplaces a direct role in supporting nearby private production. Private agglomeration then amplifies the response as private employment adjusts. I hold $\lambda_{PP} = 0.071$ fixed because the relocation responses reflect both mechanisms together. The estimated public elasticity applies to the weighted public density index in equation (9).

7.3 Policy Counterfactual

I simulate the relocations completed between 2015 and 2024 as a single shock to the baseline economy. Let $n_{rs} \geq 0$ denote positions transferred from origin s to destination r. The net change in public workplace employment is $\Delta\bar{H}_{Mj}^G = \sum_s n_{js} - \sum_r n_{rj}$, giving

$$\bar{H}_{Mj}^{G,\text{cf}} = \bar{H}_{Mj}^{G,0} + \Delta\bar{H}_{Mj}^G, \tag{17}$$

$$\sum_j \Delta\bar{H}_{Mj}^G = 0. \tag{18}$$

National public and private worker masses remain fixed. A county that both sends and receives public positions is classified by its net position.

Each transferred position carries its origin's baseline office requirement:

$$\bar{L}_j^{G,\mathrm{cf}} = \bar{L}_j^{G,0} + \sum_s n_{js}\ell_s^{G,0} - \sum_r n_{rj}\ell_j^{G,0}. \tag{19}$$

Vacated offices return to the space available for private production and housing. Arriving offices generally occupy the destination's existing stock. For genuinely new state offices completed by 2024, I add the corresponding space to the destination's total stock as well as its public office occupancy. Office requirements follow the origin, while pay follows the destination's equilibrium wage. The government continues to value every office at its commercial rental equivalent, and the tax rate adjusts to balance the budget.

I re-solve for wages, residential and commercial rents, private workplace and residence choices, public residence choices, floor-space use, productivity, amenities, and the tax rate. Productivity and amenity fundamentals, taste scales, rent wedges, land, travel times, and national worker masses remain at their baseline values. Existing floor-space supply is fixed apart from the documented additions. The comparison describes the difference between the relocation equilibrium and the 2011 baseline equilibrium without the relocations.

7.4 Counterfactual Results

Receiving counties gain private workplaces alongside the relocated public jobs: private workplace employment rises by 2.0% relative to the no-relocation baseline. Employment among residents of these counties rises by only 0.11% across both sectors. Receiving places therefore gain workplace activity without a comparable increase in employment among residents. Commuting connects the additional jobs to a wider residential workforce, consistent with the distinction between jobs and residence in the worker evidence.

Private firms in receiving counties become more productive and expand production. Private productivity rises by about 1.0%, and private output rises by 2.8%. Holding fundamentals fixed, the productivity response is

$$\Delta \log A_j^P = \lambda_{PP}\Delta \log \Upsilon_j^P + \lambda_{PG}\Delta \log \Upsilon_j^G. \tag{20}$$

The relocated public workplaces increase nearby public density, and the induced private expansion adds private agglomeration. These productivity effects allow the receiving private sector to grow even though the agencies also occupy floor space.

The receiving private expansion increases demand for space. Private commercial floor space rises by 1.5%, while residential floor space falls by 1.0%. Commercial and residential rents rise by about 1.3% and 1.2%, respectively, alongside a 0.9% increase in nominal wages. The gains in private activity thus come with higher housing costs. At the origins, moving offices releases space that private firms or households can use. This competition for space is the counterweight to productivity spillovers emphasized in the introduction.

The gains at the destinations leave national aggregates nearly unchanged. Private output in Germany falls by 0.035%, while private workers' expected utility rises by

0.017%. The concentration of private output across counties, measured by the Herfindahl–Hirschman index, falls by 0.35% relative to its baseline value. The program spreads activity more widely with a small aggregate output cost. Private workers' living standards reflect the combined adjustment of disposable wages, housing costs, amenities, and commuting, rather than output alone.

Table 5. Private activity and aggregate effects of the relocations

Outcome	Change (%)
Receiving counties	
Private workplace employment	2.01
Employed residents, both sectors	0.11
Private productivity	0.97
Private output	2.80
Germany	
Private output	−0.035
Private workers' expected utility	0.017
Concentration of private output (HHI)	−0.35

Notes: Changes compare the equilibrium with relocations completed by 2024 with the no-relocation baseline at $\widehat{\lambda}_{PG} = 0.364$. Employment and output changes refer to group totals. Productivity is weighted by baseline private workplace employment. Resident employment includes private and public employed residents. HHI denotes the Herfindahl–Hirschman index of county shares in national private output; its change is relative to the baseline index.

7.4.1 Channels

Public-to-private productivity spillovers account for most of the receiving employment gain. When I set $\lambda_{PG} = 0$, receiving private employment changes by -0.02% instead of rising by 2.0%. Private agglomeration amplifies this gain: setting $\lambda_{PP} = 0$ leaves a receiving increase of 1.7%. The relocation of offices restrains private employment at the destinations and supports it at the origins, by about 0.007 percentage points in each case. New state-built floor space adds about 0.002 percentage points to the receiving response. I measure each channel by comparing the full equilibrium with the equilibrium in which that channel is switched off; these comparisons allow the other channels to adjust together.

7.4.2 Allocation of Public Jobs

The small national output loss also depends on where the public jobs go. I rank counties in the policy state by the effect on national private output of moving the same marginal bundle of public positions from Munich to each county in turn. I then compare the actual allocation with an allocation to the highest-ranked destinations. Both keep the public positions and their origins unchanged and use the same number of destination counties. In the alternative, I distribute arrivals and their transferred office space evenly among

the selected destinations. I exclude new state-built floor space from both comparisons to compare the allocation of public jobs across destinations.

Choosing destinations with the highest marginal contribution to private output cuts the national output loss by about two thirds, from 0.035% under the actual destinations to 0.012%. Private workers' expected utility rises by 0.024% under this allocation, compared with 0.017% under the actual allocation. Destination choice can therefore preserve the regional redistribution of public jobs while reducing its aggregate output cost. The gains from public density and the competition for space vary across locations, giving the government scope to improve the allocation of its workplaces.

8 Conclusion

Relocating public employment can bring private jobs to lagging places. Studying a program that moves public agencies from thriving to lagging municipalities, I find that private workplace employment in receiving municipalities rises by 4.3% after five years. Sending municipalities show no evidence of offsetting private employment losses. The sector responses differ across the two ends of the program: food and beverage services expand at the destinations, while knowledge services expand at the origins.

Public jobs move much more than the original public workforce. Few incumbents change residence and take jobs at receiving agencies, while many retain their residence and continue working in public administration. Receiving agencies recruit almost their entire workforce from outside the original sending cohort. They become more female, while the female share of private employment in receiving municipalities decreases.

The quantitative spatial model connects these local responses to national private production and private workers' living standards. Public workplaces generate productivity spillovers, and public offices compete with private firms and households for floor space. The relocations raise receiving private output by 2.8%, while national private output falls by only 0.035% and private workers' expected utility rises by 0.017%. Private production becomes less concentrated across counties.

The location of public jobs shapes the aggregate cost of regional redistribution. Assigning the same jobs to destinations with the highest marginal contribution to private output reduces the national output loss by about two thirds. Governments can therefore improve the allocation of their workplaces by considering both the private activity that public jobs support and the space that public offices occupy.

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Appendix

Appendix A documents institutions and costs. Appendix B describes the data and worker samples. Appendix C gives the empirical specifications and diagnostics. Appendix D derives the model. Appendix E reports additional empirical specifications. Appendix F reports residence and local employment results.

Appendix A: Institutional Setting

Public Employment and Pay

Germany's June 2024 public-service workforce comprised approximately 527,000 federal employees, 2.67 million state employees, 1.80 million municipal employees, and 381,000 employees of social insurance institutions (Destatis 2025; Destatis 2026). In the World Bank series, the public wage bill remained close to one-sixth of total public expenditure from 2000 to 2019. Its share declined in 2020 as total spending rose during the pandemic (World Bank 2024).

Bavaria's direct state administration employed 311,580 people in 2014, including 213,090 civil servants and judges. The count includes full-time and part-time staff and excludes municipal employers and legally separate public institutions (Statistical Offices 2014; Bavarian Statistical Office 2014). I use this population to benchmark the policy's intended size.

	Monthly gross base pay (€)	
Grade (published steps)	First step	Last step
E9a (1, 6)	3,520.10	4,490.04
A9 (2, 10)	3,294.99	4,108.12
E13 (1, 6)	4,629.74	6,580.44
A13 (5, 11)	5,247.58	6,168.25

Table 6. Contractual and civil-service pay schedules, effective February 1, 2025. Amounts are full-time monthly base pay at each grade's first and last published steps. The endpoints do not represent equal experience across employment statuses. Annual special payments and other compensation are excluded. Source: LfF (2025b) and LfF (2025a).

Bavarian budget law pairs E9 with A9 and E13 with A13 for the purpose of comparing budgeted posts; this is not a rule assigning employees to grades (Freistaat Bayern 2024, Art. 6a). E9a covers administrative duties accessible without a university degree. In the non-graduate civil-service career, workers normally enter at A6 or A7 and reach A9 through promotion. A9 can also be an entry grade for the degree-based career track. E13 and A13 illustrate master's-level work rather than all university-qualified employment (TdL 2025; Freistaat Bayern 2026).

A non-graduate who reaches A9 through promotion will already have accumulated service. The first published A9 step therefore does not provide an equal-experience comparison with an E9a entrant. Civil servants and contractual employees also face different pension and insurance arrangements, so gross base pay alone does not determine

their relative take-home pay (Freistaat Bayern 1992). A net-pay comparison would require a specified career history, tax situation, and health-insurance arrangement. For the policy, the common feature is that a given grade and experience step carries the same scheduled base pay across Bavarian municipalities.

Program Boundaries and Position Counts

Phase	Administrative designation	Coverage in this paper
I	*Regionalisierung von Verwaltung*	Included
I	*Strukturkonzept – Chancen im ganzen Land*	Included
II	*Behördenverlagerung Bayern 2030*	Included; implementation under way
Other measures	Individual projects outside phases I and II	Not included

Table 7. The two phases of the relocation program. The empirical sample covers projects with an observed opening within the outcome-data window. Source: Bavarian Ministry of Finance (2025) and Kaltenhauser (2023).

Phase	Intended jobs	Paid training positions
I	2,511	730
II	2,685	440
Total	5,196	1,170

Table 8. Intended final staffing across both phases. Source: Bavarian Ministry of Finance (2026, pp. 6, 16). These are the government's final staffing figures, not current employment or the number of incumbents who move. Training positions include remunerated public-service degree programs.

The intended jobs equal 1.668% of the pre-policy direct state workforce (5,196/311,580). Including paid training positions gives 2.043% (6,366/311,580). Section 2 reports the jobs-only ratio to keep the two categories distinct.

Geographical Targeting

The local reconstruction identifies 829 municipalities through the district index and 392 through the municipal index. Their union contains 885 of Bavaria's 2,056 municipalities, with overlap counted once. The reconstruction uses an index value of 90 or below; the government describes eligibility as below 90, and published scores are rounded to one decimal place. Service-center designations and ministerial site selection operate in addition to these index-based eligibility criteria. Figure 1 illustrates these geographical selection layers.

Weight	Indicator	District reference period
30%	Projected population change	2010–2030
30%	Unemployment rate	2007–2011
10%	Employment density	2007–2011
20%	Disposable household income per resident	2005–2009
10%	Net migration of ages 18–29 per 1,000 in that age group	2006–2010

Table 9. Composition of the structural index. Weights and periods refer to the district index underlying the 2014 expansion of eligibility. The Bavarian mean is 100 and the expanded threshold is 90. Municipal qualification uses a comparable index with municipal proxies, including taxable income. Source: Bavarian Ministry of Finance (2015) and StMWi (2014).

Implementation Costs

Reported category	Amount (€)
Budgeted temporary staffing costs, 2016–2022	16,623,849.00
Construction expenditure, four sites	74,923,320.64
Renovation expenditure	3,708,003.37
Standardized moving-cost estimates, 2015–2021	450,713.00
Standardized rental-cost estimates, 2015–2022	18,188,015.00
Sum of reported categories	113,893,901.01

Table 10. Reported cost categories for completed relocation projects. Source: Kaltenhauser (2023, pp. 13–14), response dated May 23, 2023. The sum combines actual construction and renovation spending, budgeted costs of temporary posts, and standardized estimates. Categories cover different periods. It is neither a measure of total cash expenditure nor the full program's lifetime net cost.

For three tax-office relocations, the same report provides construction spending and the number of jobs established by completion. Höchstädt an der Donau accounts for €9,712,273.52 and 53 jobs, Waldmünchen for €4,358,424.71 and 21 jobs, and Bad Kötzting for €2,691,643.64 and 20 jobs (Kaltenhauser 2023, pp. 8, 13). Summing spending and jobs across these sites gives €16,762,341.87/94, or €178,323 per public job, rounded to €180,000 in the main text. This construction-only ratio excludes other transition costs, savings at origins, and the value of the resulting assets. It does not measure the fiscal cost per additional private job.

Regional investment subsidies under the German GRW program provide a further benchmark. Using regional variation in subsidy rates, Grunau et al. (2025) estimate a cost of about €25,000 per additional job. Their causal employment denominator differs from the public positions accommodated by the three relocation projects.

For Northvolt's planned battery plant in Heide, the European Commission approved €700 million in direct grants and €202 million in guarantees in January 2024 (European Commission 2024). The government announced 3,000 direct jobs (German Federal Government 2024). Dividing the grants by planned direct employment gives €233,333

per job, rounded to €230,000 in the main text. This calculation excludes guarantees, whose face value is not a grant expenditure, and projected indirect jobs.

For Intel's planned semiconductor plant in Magdeburg, the federal government proposed €9.9 billion in support in June 2023, subject to European Commission approval, alongside about 3,000 direct jobs (German Federal Government 2023). This implies €3.3 million per planned direct job, excluding temporary construction and projected indirect employment. Intel cancelled the project in July 2025 (Intel 2025). This ratio therefore describes the original subsidy offer, not a realized cost per job.

Both firm-specific benchmarks measure announced support per planned direct job, not a causal cost per additional job. All amounts are in nominal euros at their respective source dates. A cost-effectiveness comparison would require net policy costs and additional employment measured over comparable horizons.

Structured News Search

I conducted a structured German-language news search in Nexis Uni using the search string

> **Behördenverlagerung OR Behördenumzug OR "Umzug von Behörden" OR "Umzug vom Amt" OR "Amtsumzug"**.

I restricted the search period to articles published from 1991 onward and limited the sample to news articles, excluding private-company publications, individual webpages, and legal texts.

The query yielded 1,531 results. I identified news reports on 22 federal and 87 state agencies between 2005 and 2032. In total, the reports cover 13,395 relocated jobs. The largest number of relocations occurs in 2019, with 5,843 jobs, followed by 2030, with 2,300 jobs, and 2021, with 1,070 jobs. Table 11 reports the number of relocated jobs and study places by destination state, and Table 12 lists all sending and destination cities.

The search identifies both single-agency relocations and broader relocation programs. The three main programs are the federal program for coal regions, the Bavarian *Heimatstrategie*, and the Saxon *Lausitzprogramm 2038*. I use this search to document the broader institutional context of public employment reallocation in Germany.

State	N of Relocated Jobs	N of Relocated Study Places
Bavaria	9,681	1,730
Brandenburg	510	
Hesse	100	
Saxony	2,214	
Saxony-Anhalt	850	
Thuringia	40	

Table 11. Number of public-sector jobs and study places relocated to each state according to the Nexis Uni news search.

Sending Cities	Destination Cities
Ammersee	Amberg, Ansbach, Augsburg
Berlin, Bonn	Bad Kissingen, Bad Königshofen im Grabfeld,
Bad Kissingen	Bad Kötzting, Bad Neustadt an der Saale, Bad Reichenhall,
	Bad Steben, Bautzen, Berlin, Bischofswerda, Borna
	Cottbus
Dresden	Dessau, Dinkelsbühl, Dresden
Erlangen, Eschborn, Essen	Ebern, Eichstätt
	Freital, Freyung, Fürstenfeldbruck, Fürth
Großenhain	Garmisch-Partenkirchen, Gemünden am Main, Gera,
	Görlitz, Grafenau, Gunzenhausen
Herrsching a. Ammersee	Halle (Saale), Höchstädt a.d.Donau, Hof
	Ingolstadt
	Kaufbeuren, Kemnath, Kitzingen, Kronach, Kulmbach
	Lauterbach, Leipzig, Lohr am Main
München	Marktredwitz, Meißen, Merseburg, Mittenwald
	Mühldorf a. Inn, Münchberg
Nürnberg	Nabburg, Neustadt an der Aisch, Nördlingen, Nürnberg
	Obernburg am Main, Oberviechtach
	Pegnitz
Regensburg	Regen, Rosenheim
Starnberg	Schwandorf, Schweinfurt
	Tirschenreuth, Traunstein
	Vilshofen an der Donau, Vohenstrauß
Würzburg	Waldkirchen, Waldmünchen, Waldsassen, Weiden in der Oberpfalz
	Weißenburg i. Bay., Weißwasser, Windischeschenbach, Wunsiedel
	Zittau, Zwiesel

Table 12. Sending and destination cities identified in the Nexis Uni news search, listed alphabetically. Some cities, such as Berlin and Dresden, appear both as sending and destination cities.

Appendix B: Data

Relocation records and the agency register

The project records distinguish intended staffing, reported implementation dates, and construction milestones. Source links document the evidence behind each entry. A program entry may represent an entire agency, a branch, or part of an existing organization. Neither an agency's continued presence at its origin nor the absence of a separately named destination in the register establishes that no employment was relocated.

I reconcile the 2014, 2023, and 2026 register editions first by agency identifier. Among unmatched records, I use normalized agency names that occur only once in each edition and then normalized addresses that occur only once in each edition. Normalization removes differences in capitalization, punctuation, and spacing. I review location changes against agency and program sources, distinguishing changes within a municipality from moves between municipalities. Shared addresses and renamed units require particular care. The register records administrative units and locations; it is not an employment census. Its coverage includes state agencies and schools, as well as other institutions with state personnel. These cross-year links remain separate from the establishment identifiers in the IAB data.

IAB descriptive samples and candidate establishments

Table 1 reports the treated municipalities in the one-to-one matched samples used for the municipal analysis. The comparison uses controls outside Bavaria. Treatment dates come from the official year where available and otherwise the recorded completion year. The sample retains *Regionalisierung von Verwaltung* projects with a treatment date by 2024 and excludes projects that are still being planned. The receiving sample contains 34 municipalities and the sending sample six. The table reports municipal means, rather than pooled worker totals.

For the worker analysis, workplace postcodes identify public-administration establishments near program origins and destinations. An establishment is classified by its identifier and postcode. A postcode can appear among both origins and destinations and is retained in both sets. Establishments observed in both 2010 and 2013 provide the baseline distribution of employment changes over that interval. I calculate its standard deviation separately for the origin and destination groups. Among continuing establishments, the change from 2014 to 2024 must be at least twice the destination-group standard deviation to qualify as a candidate expansion, or at most minus the origin-group standard deviation to qualify as a candidate contraction. Establishments observed only in 2024 qualify as candidate openings, and those observed only in 2014 as candidate closures. These rules define the establishments used in the descriptive worker comparisons.

The source cohort consists of workers employed at candidate sending establishments in 2014. Residential movers and commuters are distinguished among those employed at candidate receiving establishments in 2024 using changes in recorded residence. Employees at candidate receiving establishments who were not in the source cohort are additional employees relative to that cohort, not necessarily new labor-market entrants. Workers no

longer observed in regular employment in 2024 are very likely retirees.

The agency-profile comparisons in Figures 4 and 5 use closed establishments in 2014 and opened establishments in 2024. The worker classification also includes substantial contractions and expansions, as defined above. The model's Census inputs select total workplace and residence employment and the industry O components, keeping published totals separate from their component breakdowns.

Appendix C: Empirical Methodology

Matching and comparison groups

I construct receiving and sending matches separately. Let X_i contain the matching characteristics described in Section 5. The squared Mahalanobis distance between a treated municipality and a potential control is

$$d_{ij}^2 = (X_i - X_j)' S^{-1} (X_i - X_j), \tag{21}$$

where S is the covariance matrix of the matching variables. Nearest-neighbor matching without replacement selects one control for each treated municipality. The covariance adjustment accounts for the different scales and correlations of the matching variables.

The main comparison pool excludes Bavarian municipalities to limit exposure to policy spillovers within the state. It also excludes municipalities in 20 non-Bavarian counties selected for their commuting links to Bavaria. Figure 21 shows the top and bottom 20 non-Bavarian counties by their share of commuters to Bavaria in 2013. The top-20 group defines the additional exclusion; the bottom-20 group is shown for comparison and does not define the donor pool.

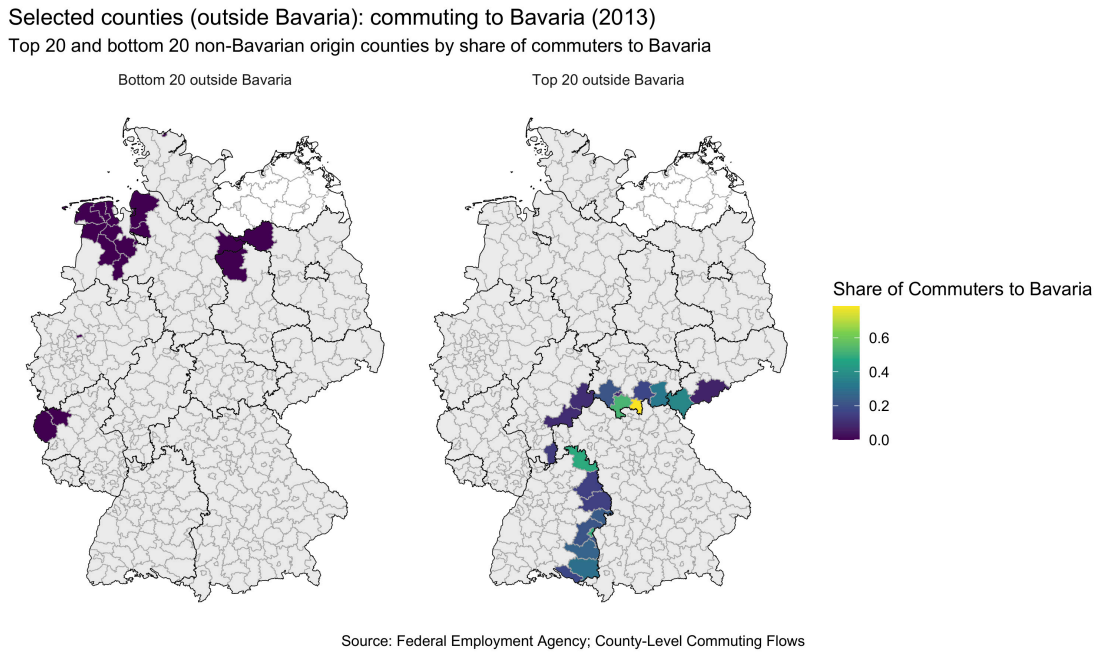


Figure 21. Commuting links used to restrict the comparison pool. Counties outside Bavaria with the lowest and highest shares of commuters to Bavaria in 2013. Municipalities in the top-20 counties are excluded from the main control pool, in addition to municipalities within Bavaria. Source: Federal Employment Agency county commuting flows.

For the auxiliary specifications that match on pre-policy levels and trends, Table 13 reports absolute standardized mean differences after matching with one and three outside-Bavaria controls. The receiving and sending columns use their respective treated-group

standard deviations. Figure 22 also shows the unmatched comparison. These statistics describe balance on the matching variables; the event-study leads provide separate evidence on pre-treatment outcome dynamics.

	Receiving		Sending	
Covariate	1:1	3:1	1:1	3:1
Population, 2014	0.001	0.060	0.058	0.197
Residence employment, 2014	0.011	0.064	0.019	0.221
Unemployment, 2014	0.113	0.005	0.318	0.060
Income per taxpayer, 2014	0.028	0.076	0.330	0.322
Population aged 18–30, 2014	0.015	0.077	0.021	0.210
Population change	0.099	0.001	0.445	0.256
Residence-employment change	0.089	0.085	0.051	0.096
Unemployment change	0.023	0.047	0.737	0.330
Income change	0.111	0.102	0.348	0.040
Population aged 18–30 change	0.043	0.074	0.962	0.681

Table 13. Balance after Mahalanobis matching with outside-Bavaria controls. Entries are absolute standardized mean differences for one and three controls per treated municipality, corresponding to Figure 22. Levels refer to 2014; changes are log differences from 2009 to 2014, except income per taxpayer, which uses 2010–2014.

I examine matching ratios of three controls per treated municipality. These checks vary the available comparisons while retaining the distinction between receiving and sending locations. I also examine sending-location estimates without Munich, whose scale differs sharply from that of the other origins.

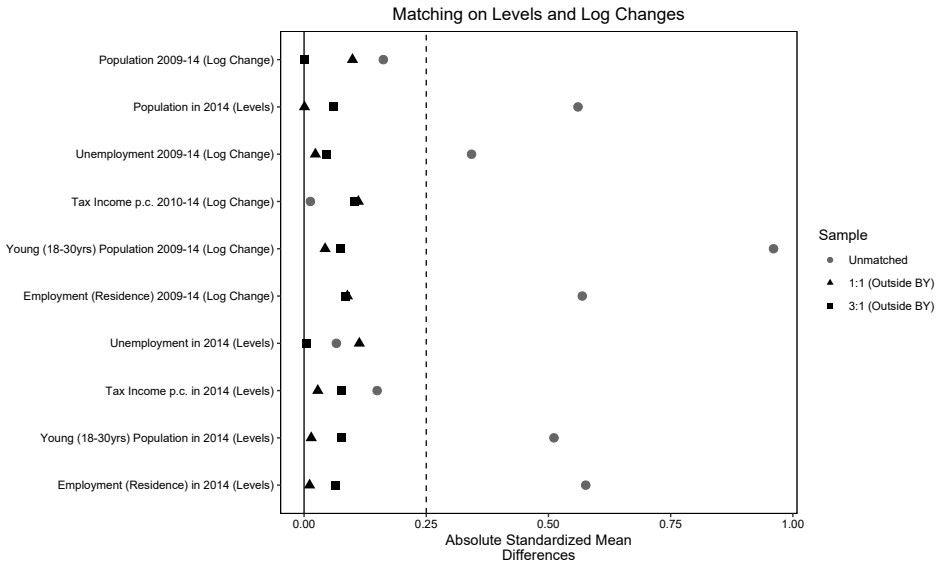
Treatment cohorts and repeated relocations

The baseline treatment year is the first recorded relocation for a municipality among the projects included in the analysis. The last-treatment year is the latest such date. Both use the official treatment year when available and otherwise the recorded completion year. Repeated dates at a location therefore describe further exposure to the policy, rather than additional independent treated municipalities. Table 14 gives the first- and last-treatment distributions underlying Figure 6.

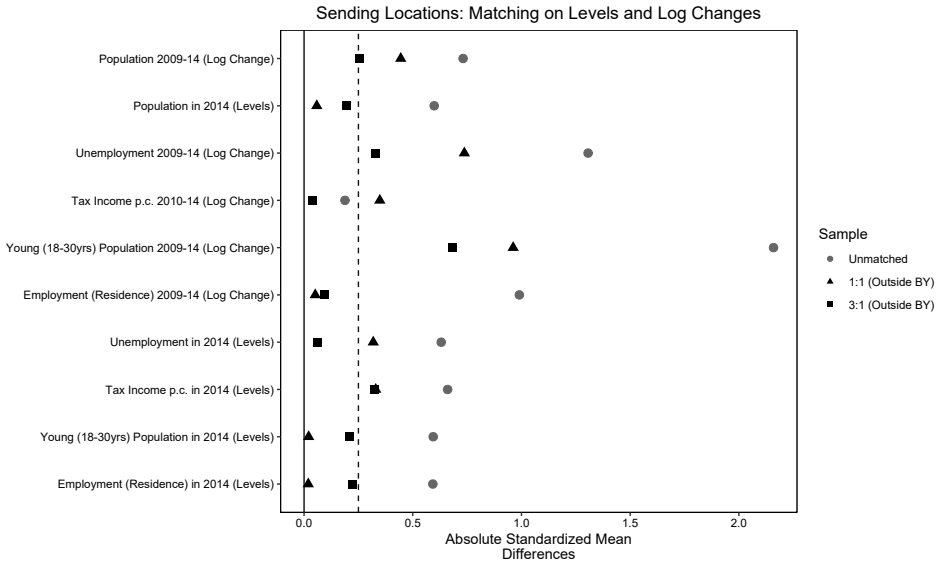
Event-time aggregation and inference

In the Sun–Abraham specification, each cohort's event-time effects are measured relative to the year before first treatment. I aggregate cohort-specific coefficients at common event times using the estimator's cohort weights. Cohorts contribute only where their outcomes are observed, so the composition of the aggregate effect can vary across event times. The municipal panel is allowed to be unbalanced. Municipality-clustered standard errors allow serial correlation within municipalities.

The Callaway–Sant'Anna specification uses never-treated municipalities as controls. For cohort g, the group-time effect compares its change from $g-1$ to t with the corresponding



(a) Receiving municipalities



(b) Sending municipalities

Figure 22. Covariate balance for Mahalanobis matching on 2014 levels and pre-policy log changes. The figures compare the unmatched pool with one-to-one and three-to-one outside-Bavaria matches. Symbols distinguish the three comparisons. The horizontal axis reports absolute standardized mean differences.

	Receiving		Sending	
Year	First	Last	First	Last
2015	3	1	2	1
2016	15	9	2	2
2017	7	8	1	1
2018	3	5	1	1
2019	0	1	0	0
2020	3	4	0	0
2021	2	2	2	2
2022	1	3	0	0
2023	1	2	0	1
2024	0	0	0	0
Municipalities	35	35	8	8

Table 14. First and last treatment years in the common matching roster. Each municipality appears once in each timing distribution. First and last refer to the earliest and latest recorded treatment among included projects, using the official year when available and the completion year otherwise.

change in the never-treated group:

$$\begin{aligned} ATT(g,t) = E[Y_{it} - Y_{i,g-1} \mid G_i = g] \\ - E[Y_{it} - Y_{i,g-1} \mid G_i = \infty]. \end{aligned} \tag{22}$$

The implementation uses a universal pre-treatment base period, permits an unbalanced panel, and aggregates group-time effects into dynamic effects. Bootstrap inference provides simultaneous confidence bands. This specification is estimated within the common matching design; it does not impose the matched-pair-by-year fixed effects of the Sun–Abraham regression.

Synthetic difference-in-differences for sending municipalities

Because only five municipalities are net senders, I complement the matched design with synthetic difference-in-differences (Arkhangelsky et al. 2021). I estimate one model per sending municipality. Donors are never-treated municipalities outside Bavaria with at least 10,000 inhabitants in 2014. As in the matched design, I exclude districts with strong commuting ties to Bavaria, every municipality named in the relocation records, and the municipalities of the other relocation programs listed in Appendix A. The pre-treatment window is 2008–2014 for all sending municipalities; years between 2015 and a later first relocation are not used. Unit weights reproduce each sending municipality's pre-treatment outcome path up to a constant. Time weights select the pre-treatment years that best predict the donors' post-treatment outcomes; in practice they fall almost entirely on 2014, so effects are measured relative to the last year before the program. Standard errors come from placebo replications. In each replication, every sending municipality's timing is assigned to one of the 100 donors closest to it in 2014 population, and the models are re-estimated without the drawn donors. Placebo units of similar size matter because small municipalities have much noisier outcomes than large cities.

The donor weights match pre-treatment outcomes but not the characteristics used in the matching. To condition on these characteristics, I follow the two-step procedure of Kranz (2021). The covariates Z_i are the matching variables: the logs of 2014 population, unemployment, residence employment, population aged 18–30 and income per taxpayer; the 2009–2014 log changes in population, unemployment, residence employment and population aged 18–30; and the 2010–2014 log change in income per taxpayer. Because Z_i is fixed over time, it can only affect outcomes through year-specific coefficients. In the first step, I estimate

$$Y_{it} = \alpha_i + \delta_t + \sum_{s \neq 2014} \mathbb{1}[t = s]\, Z_i' \gamma_s + \varepsilon_{it} \tag{23}$$

by OLS on all untreated municipality-years: donor municipalities in every year and sending municipalities before their first relocation. Using only untreated observations keeps the relocations themselves out of the estimated covariate effects. In the second step, I remove the fitted covariate component,

$$\tilde{Y}_{it} = Y_{it} - \sum_{s \neq 2014} \mathbb{1}[t = s]\, Z_i' \hat{\gamma}_s, \tag{24}$$

and apply synthetic difference-in-differences to $\tilde{Y}_{it}$ with the same donors, time windows and placebo procedure. The adjustment allows, for example, larger municipalities to follow different trends than smaller ones. The placebo standard errors take $\hat{\gamma}_s$ as given and do not reflect estimation error from the first step (ibid.).

As a falsification test, I backdate treatment to 2012 for every sending municipality and estimate the same models with 2008–2011 as pre-treatment years and 2012–2014 as placebo post-treatment years. Because the analyzed relocations start in 2015, these estimates should be close to zero. The dynamic version reports the treated-minus-synthetic gap in each year relative to the time-weighted pre-treatment baseline, with pointwise placebo bands.

Appendix D: Quantitative Model

This appendix derives the worker choices and baseline inversion for the model in Section 7. It documents how wages, productivity, amenities, office requirements, housing demand, and the public budget combine to rationalize the baseline economy.

Data preparation. I harmonize the data to German county boundaries in 2011. The Census inputs measure total employment by workplace and residence. In the published files `2000X-2041` and `2000X-2045`, I select the total-commuting cell and either the all-industry total or industry O, keeping aggregates separate from their components. Industry O defines public employment through public administration, defense, and compulsory social security; its complement defines private employment. These are industry classifications. Road travel times τ_{ij} are measured in minutes, with residence first and workplace second in worker choices. In the productivity and amenity kernels, the first index is the recipient and the second the source of density. Travel links across the policy-state boundary remain in the model.

Commuting parameters. The gravity equation identifies $\nu = \epsilon\kappa = 0.0429$ per minute. Given residence i, private conditional workplace choice is

$$\pi^P_{j|i} = \frac{D_{ij} E^P_j w^\epsilon_j}{\sum_s D_{is} E^P_s w^\epsilon_s}, \qquad H^P_{Mj} = \sum_i H^P_{Ri} \pi^P_{j|i}. \tag{25}$$

To calibrate ϵ, a preliminary private-sector inversion uses observed private workplace and private resident employment to recover transformed wages $\omega^{P,\text{cal}}_j = E^P_j (w^{\text{cal}}_j)^\epsilon$ up to scale. Following Ahlfeldt, Redding, et al. (2015), I set $E^P_j = 1$, giving $w^{\text{cal}}_j = (\omega^{P,\text{cal}}_j)^{1/\epsilon}$. Matching the cross-county variance of log nominal wages at this stage gives

$$\epsilon^2 = \frac{\text{Var}_j(\log \omega^{P,\text{cal}}_j)}{\text{Var}_j(\log w^{\text{data}}_j)}, \qquad \widehat{\epsilon} = 2.486, \qquad \widehat{\kappa} = \widehat{\nu}/\widehat{\epsilon} = 0.0172.$$

Choice probabilities. Independence of the Fréchet draws gives the distribution of the best private alternative,

$$\Pr\Big(\max_{i,j} u^P_{ijo} \le u\Big) = \exp\Big(-\Phi^P u^{-\epsilon}\Big).$$

For a public worker assigned to j, the corresponding distribution is

$$\exp\Big(-[(1-t)w_j]^\epsilon \Psi^G_j u^{-\epsilon}\Big).$$

These distributions yield the probabilities and expected utilities in equations (3)–(5); expectations exist for $\epsilon > 1$. Bilateral flows satisfy

$$\sum_{i,j} N^P_{ij} = \bar{H}^P, \qquad \sum_i N^G_{ij} = \bar{H}^G_{Mj}, \qquad \sum_i H^m_{Ri} = \sum_j H^m_{Mj} = \bar{H}^m. \tag{26}$$

Private probabilities sum to one over all residence–workplace pairs. Public residence probabilities sum to one separately within each assigned workplace, preserving the government's workplace allocation.

Residential inversion. In the baseline inversion, I set $T_i^P = T_i^G = E_j^P = 1$ and recover a common effective amenity, giving $Z_i^P = Z_i^G = Z_i$. Define $\omega_j^{P,0} = (w_j^0)^\epsilon$, $\mathcal{W}_i^{P,0} = \sum_j D_{ij}\omega_j^{P,0}$, and $\Psi_j^{G,0} = \sum_r Z_r^0 D_{rj}$. The inversion jointly solves for $(\omega^{P,0}, Z^0)$ using the private workplace and total residence margins:

$$
\begin{aligned}
H_{Mj}^{P,0} &= \bar{H}^P \omega_j^{P,0} \frac{\sum_i Z_i^0 D_{ij}}{\sum_r Z_r^0 \mathcal{W}_r^{P,0}}, \\
H_{Ri}^0 &= \bar{H}^P \frac{Z_i^0 \mathcal{W}_i^{P,0}}{\sum_r Z_r^0 \mathcal{W}_r^{P,0}} + \sum_j \bar{H}_{Mj}^{G,0} \frac{Z_i^0 D_{ij}}{\Psi_j^{G,0}}.
\end{aligned} \tag{27}
$$

The common tax cancels from the employment probabilities. I scale the recovered nominal wages to the observed geometric mean of county wages. The recovered residential index gives $B_i^0 = (Q_i^{R,0})^{1-\beta}(Z_i^0)^{1/\epsilon}$. Private and public resident employment are implied separately by the two choice rules. Their earnings enter resident income and housing demand, while their combined density enters amenities.

Production and space inversion. Wages and rents recover private productivity from zero profits and private floor space from factor payments:

$$
A_j^{P,0} = \left(\frac{w_j^0}{\alpha}\right)^\alpha \left(\frac{q_j^0}{1-\alpha}\right)^{1-\alpha}, \qquad L_j^{P,0} = \frac{1-\alpha}{\alpha}\frac{w_j^0 H_{Mj}^{P,0}}{q_j^0}. \tag{28}
$$

Equation (12) gives baseline public office space. Payroll and office costs determine t^0, and bilateral flows allocate gross earnings to residences. Housing demand gives $L_i^{R,0} = (1-\beta)v_i^0/Q_i^{R,0}$. The effective stock is $L_i = L_i^{P,0} + \bar{L}_i^{G,0} + L_i^{R,0}$.

Fundamentals. For each candidate public elasticity λ, I remove the baseline density components:

$$
a_j(\lambda) = \frac{A_j^{P,0}}{(\Upsilon_j^{P,0})^{\lambda_{PP}}(\Upsilon_j^{G,0})^{\lambda}}, \qquad b_i = \frac{B_i^0}{(\Omega_i^0)^\eta}. \tag{29}
$$

Every candidate therefore rationalizes the same baseline productivity. Fundamentals remain fixed within each comparison of the baseline and relocation equilibria. The observed rent ratio $\widehat{\xi}_i = q_i^{\text{data}}/Q_i^{R,\text{data}}$ imposes the mixed-use condition $q_i = \xi_i Q_i^R$ on the residual floor-space stock. I hold this wedge fixed in the counterfactuals. A comparison with the wedge implied by wage bills, resident income, and CORINE commercial land shares provides a separate land-use check.

Baseline fiscal identity. Summing income across residences gives $\sum_i I_i = W^P + W^G$. Combining disposable income with the public budget gives $\sum_i v_i = W^P - C^G$. At baseline, the office rule implies

$$C^{G,0} = \gamma_G \frac{1-\alpha}{\alpha} W^{G,0}, \qquad t^0 = \left(1 + \gamma_G \frac{1-\alpha}{\alpha}\right) \frac{W^{G,0}}{W^{P,0} + W^{G,0}}. \tag{30}$$

With $\gamma_G = 0.25$ and $\alpha = 0.80$, the baseline office bill equals 6.25% of public payroll. After a relocation, origin-specific office requirements remain fixed, and the government budget is evaluated at counterfactual wages, rents, and occupancy.

Counterfactual accounting. Gross departures satisfy $\sum_r n_{rs} \leq \bar{H}^{G,0}_{Ms}$. Moves within a county cancel in the county workplace shock. New state offices completed by 2024 add their office requirement to both public occupancy and the total destination stock. Offices accommodated in the existing stock use space that would otherwise be available to private firms and households. Private welfare follows from the expected-utility formula:

$$\Delta \log \bar{U}^P = \epsilon^{-1} \log\left(\Phi^{P,\mathrm{cf}} / \Phi^{P,0}\right).$$

The tax factors inside Φ^P incorporate the change in disposable income.

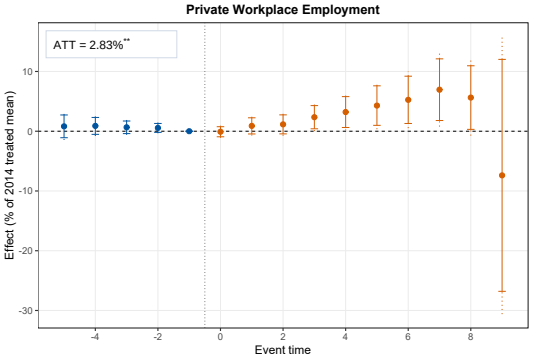
Solution. For each candidate elasticity, I invert the baseline, solve a zero-shock equilibrium, solve the relocation counterfactual, and apply the auxiliary regressions. The zero-shock solution reproduces the baseline. At each solution, I verify national worker masses, public assignments, private factor payments, housing expenditure, floor-space use, resident income, and the government budget.

Appendix E: Additional Robustness Checks

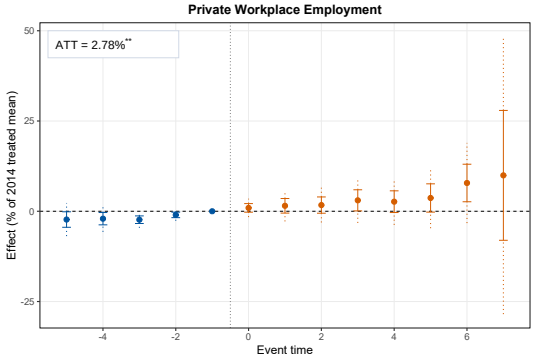
These checks change one comparison at a time and retain the distinction between employment at workplaces and at residences.

Estimator and event-time support

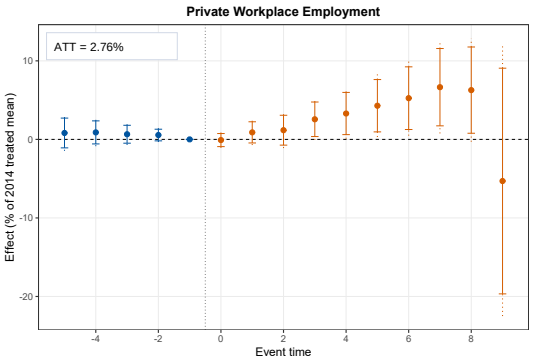
Figures 23 and 24 compare Sun–Abraham, Callaway–Sant'Anna, and two-way fixed-effect event studies for receiving and sending municipalities. The estimators aggregate treatment effects differently, and the Callaway–Sant'Anna implementation does not impose the matched-pair-by-year fixed effects of the other two.



(a) Sun–Abraham: workplace. The 2014 treated mean is 8,407 employees.

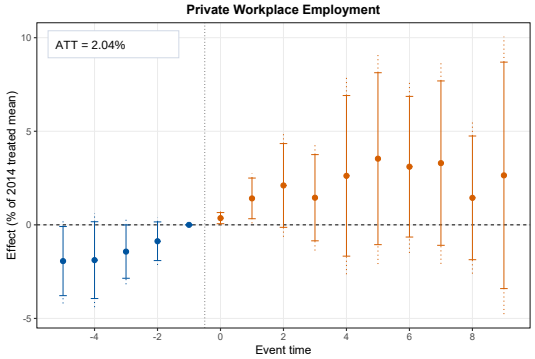


(b) Callaway–Sant'Anna: workplace. The 2014 treated mean is 8,407 employees.

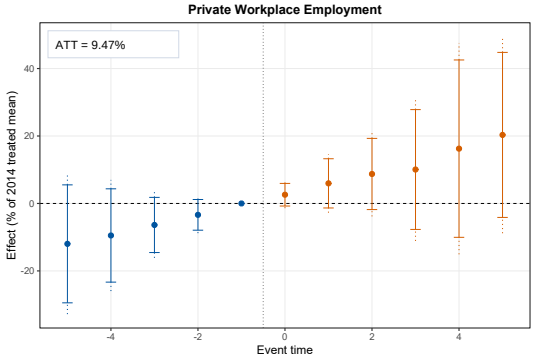


(c) TWFE: workplace. The 2014 treated mean is 8,407 employees.

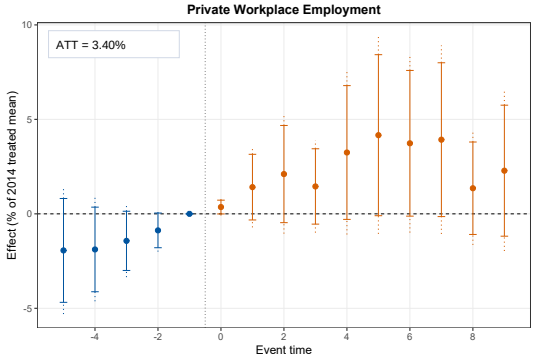
Figure 23. Receiving private workplace employment dynamics under alternative estimators, using the outside-Bavaria matching design. Table 15 reports estimates and interval conventions at event year five. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.



(a) Sun–Abraham: workplace. The 2014 treated mean is 219,230 employees.



(b) Callaway–Sant'Anna: workplace. The 2014 treated mean is 219,230 employees.



(c) TWFE: workplace. The 2014 treated mean is 219,230 employees.

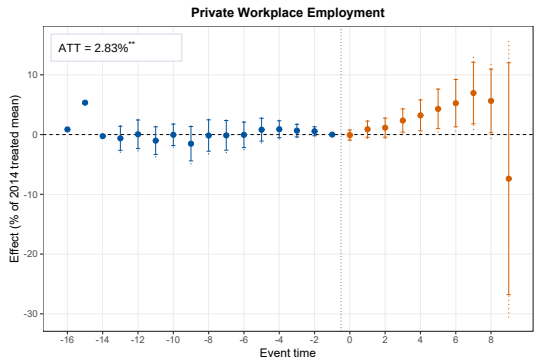
Figure 24. Sending private workplace employment dynamics under alternative estimators, using the outside-Bavaria matching design. Panel scales differ. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Municipalities	Measure	Estimator	Estimate (SE)	Interval
Receiving	Workplace	SA	4.28 (2.02)	[0.36, 8.21]
Receiving	Workplace	CS	3.69 (1.78)	[-4.64, 12.01]
Receiving	Workplace	TWFE	4.28 (2.02)	[0.36, 8.21]
Receiving	Residence	SA	3.38 (1.18)	[1.01, 5.74]
Receiving	Residence	CS	1.52 (1.86)	[-10.14, 13.34]
Receiving	Residence	TWFE	3.04 (1.35)	[0.51, 5.57]
Sending	Workplace	SA	2.27 (1.61)	[-0.88, 5.42]
Sending	Workplace	CS	20.32 (14.87)	[-797.79, 838.44]
Sending	Workplace	TWFE	2.64 (1.73)	[-0.76, 6.04]
Sending	Residence	SA	0.83 (0.67)	[-0.48, 2.13]
Sending	Residence	CS	19.94 (15.56)	[-173.47, 213.35]
Sending	Residence	TWFE	1.33 (1.53)	[-1.66, 4.33]

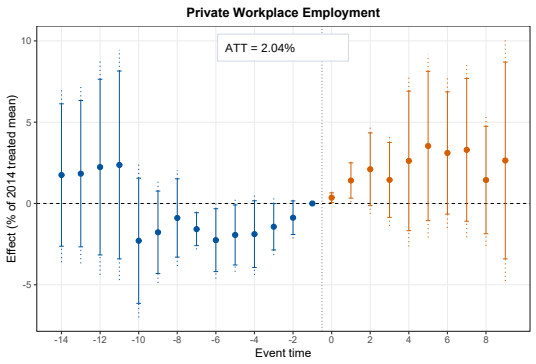
Table 15. Estimator comparison five years after first treatment. All entries are expressed as percentages of the corresponding 2014 treated mean. SA and TWFE intervals are pointwise 95% intervals based on municipality-clustered standard errors. CS intervals are bootstrap simultaneous bands, so interval widths are not directly comparable across estimators. Samples and specifications correspond to Figures 23 and 27.

For receiving workplace employment at event year five, the Sun–Abraham estimate is about 4.3% of 2014 private workplace employment and the Callaway–Sant'Anna estimate about 3.7%, but the latter simultaneous band covers both substantial losses and gains.

Figure 25 reports the full Sun–Abraham event-time ranges, which extend the windows shown above. Cohorts enter each event time only where their outcomes are observed, so remote leads and lags rest on changing cohort support.



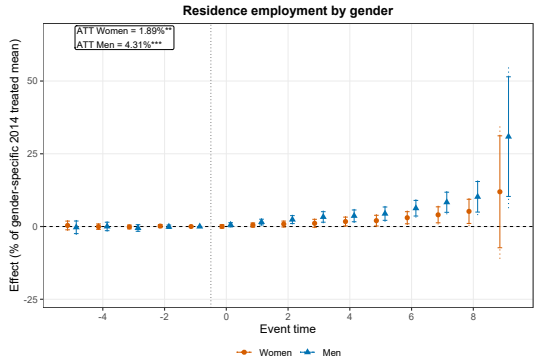
(a) Receiving: workplace. The 2014 treated mean is 8,407 employees.



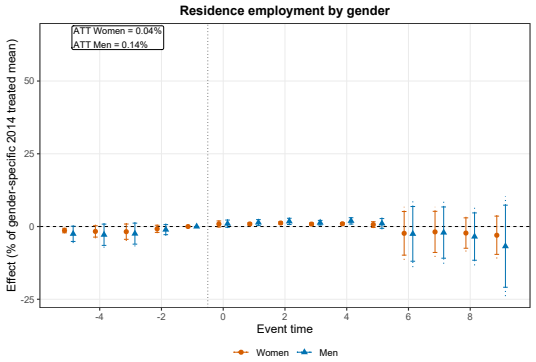
(b) Sending: workplace. The 2014 treated mean is 219,230 employees.

Figure 25. Full event-time support for private workplace employment. Sun–Abraham estimates for receiving municipalities in the left panel and sending municipalities in the right panel. The contributing cohorts change across horizons. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Appendix F: Residence and Local Employment

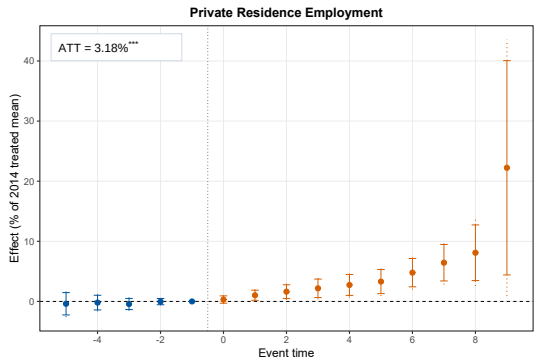


(a) Receiving. The 2014 treated means are 3,110 women and 3,141 men.

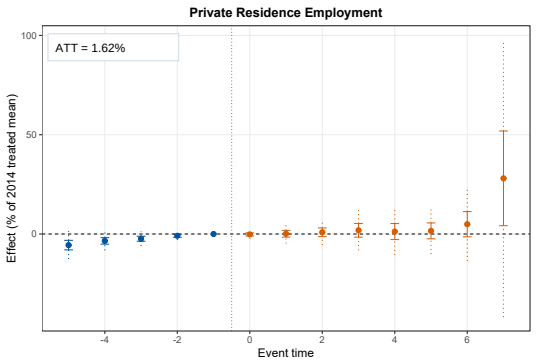


(b) Sending. The 2014 treated means are 82,782 women and 85,288 men.

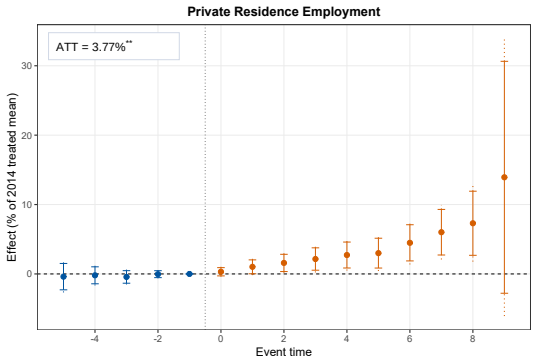
Figure 26. Residence employment by gender around the first relocation. Sun–Abraham estimates relative to matched municipalities outside the policy state. The year before treatment is the reference. Estimates and confidence intervals are expressed as percentages of the respective gender's 2014 treated mean. Solid bars show pointwise 90% confidence intervals. Dotted extensions show pointwise 95% confidence intervals. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.



(a) Sun–Abraham: residence. The 2014 treated mean is 5,920 employees.

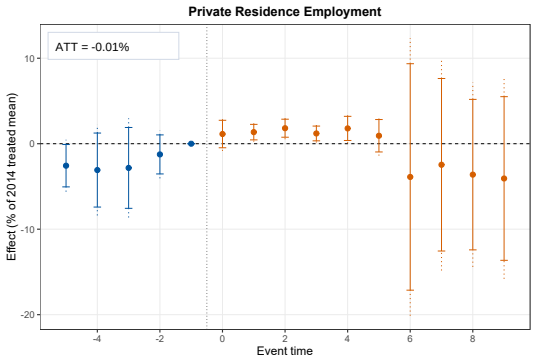


(b) Callaway–Sant'Anna: residence. The 2014 treated mean is 5,920 employees.

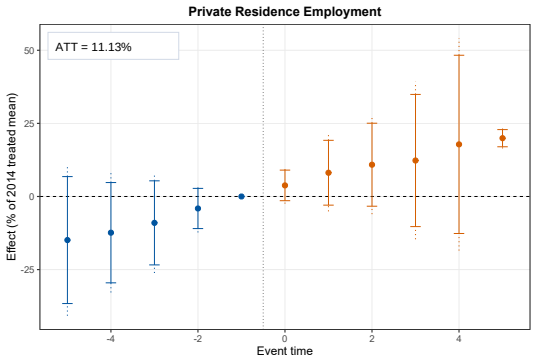


(c) TWFE: residence. The 2014 treated mean is 5,920 employees.

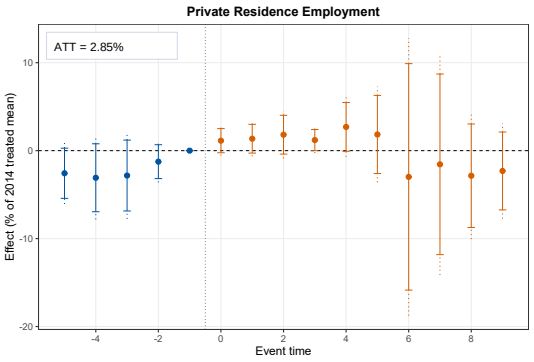
Figure 27. Receiving private residence employment dynamics under alternative estimators, using the outside-Bavaria matching design. Table 15 reports estimates and interval conventions at event year five. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.



(a) Sun–Abraham: residence. The 2014 treated mean is 161,704 employees.

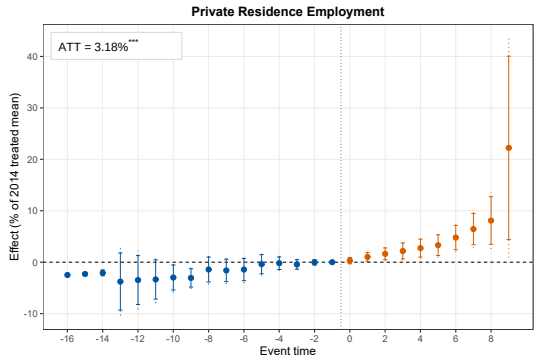


(b) Callaway–Sant'Anna: residence. The 2014 treated mean is 161,704 employees.

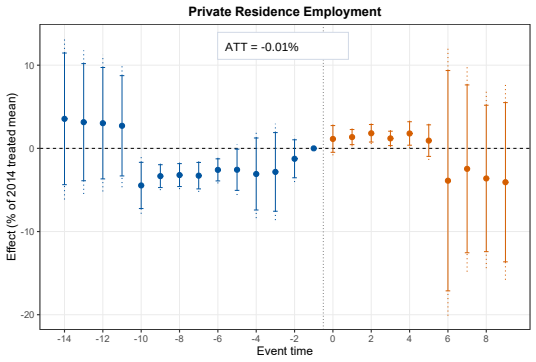


(c) TWFE: residence. The 2014 treated mean is 161,704 employees.

Figure 28. Sending private residence employment dynamics under alternative estimators, using the outside-Bavaria matching design. Panel scales differ. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.



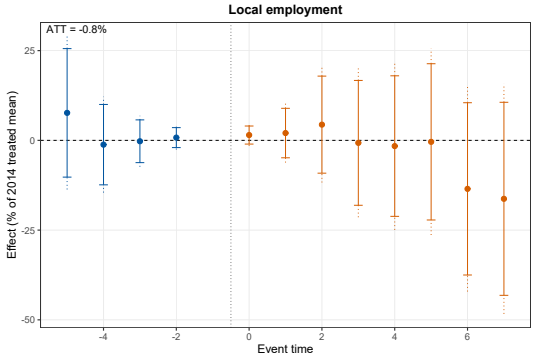
(a) Receiving: residence. The 2014 treated mean is 5,920 employees.



(b) Sending: residence. The 2014 treated mean is 161,704 employees.

Figure 29. Full event-time support for private residence employment. Sun–Abraham estimates for receiving municipalities in the left panel and sending municipalities in the right panel. The contributing cohorts change across horizons. Estimates and confidence intervals are expressed as percentages of the corresponding 2014 treated mean. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Solid bars show 90% intervals. Dotted extensions show 95% intervals.

Figure 30 reports event studies for sending locations without Munich using total employment from municipal statistics, which includes public and private employees.



(a) Local employment. The 2014 treated mean is 33,674 employees.

Figure 30. Sending-location employment responses excluding Munich. Sun–Abraham estimates expressed as percentages of the 2014 treated mean using one-to-one outside-Bavaria matches on levels and pre-policy trends. Employment includes public and private employees. Solid bars show 90% intervals. Dotted extensions show 95% intervals.